

DATA ANALYTICS

02

Statistics and Probability

Turn raw numbers into confident business decisions — master the statistical thinking every analyst needs.

DURATION	LEVEL	MODULE	UPDATED
19 Sessions	Beginner to Intermediate	Module 2	June 2026

IN THIS MODULE

01	Introduction to Statistics (Descriptive vs Inferential)
02	Types of Data & Measurement Scales
03	Measures of Central Tendency (Mean, Median, Mode)
04	Measures of Dispersion (Range, Variance, Std Dev)
05	Percentiles, Quartiles & Boxplots
06	Probability Basics
07	Conditional Probability & Bayes' Theorem
08	Probability Distributions (Discrete)
09	Probability Distributions (Continuous)
10	Sampling Techniques
11	Central Limit Theorem (CLT)
12	Hypothesis Testing Concepts
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14	Chi-Square Test
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- 18 Case Study 1 — E-Commerce Sales Analysis (Excel)
- 19 Case Study 2 — Customer Churn Analysis (Excel)

SESSION

01

60 MIN · FUNDAMENTALS

Introduction to Statistics — Descriptive vs Inferential

◆ Learning Objectives

- Define statistics and articulate its role in data-driven business decisions
- Distinguish descriptive from inferential statistics with examples
- Differentiate population from sample, and parameters from statistics
- Identify the type of statistics applied in three different business scenarios
- Demonstrate sampling error by comparing sample and population means in Excel

01 What is Statistics?

Statistics is the science of collecting, organising, analysing, and interpreting numerical data to support decisions. A kirana owner in Maninagar tracking daily sales and a bank analyst forecasting loan defaults both use statistics — at different scales and complexity. Without statistics, data is just noise.

CONCEPT — Two Branches of Statistics

Descriptive statistics summarise the data you already have — average monthly revenue, top-selling SKU, pass rate per batch. Inferential statistics use a sample to draw conclusions about a larger population — for example, surveying 500 Surat textile buyers to predict the behaviour of 50,000.

Descriptive

Summarises existing data. No predictions. Examples: mean score of 200 students, most enrolled course at TOPS, pass rate per batch.

Inferential

Draws conclusions beyond the data using probability. Example: testing if a new curriculum improves scores for all future batches.

02 Population, Sample, Parameters and Statistics

A population includes every individual in the group you want to study. A sample is a smaller, representative subset. Because studying an entire population is expensive and often impossible, analysts almost always work with samples and use inferential statistics to generalise findings.

Term	Symbol	Refers To	Example
Population	N	Entire group of interest	All Class 12 students in Gujarat 2024

Term	Symbol	Refers To	Example
Sample	n	Chosen subset	500 students selected for a skill survey
Parameter	μ, σ	Fact about population	True average marks of all 5 lakh students
Statistic	$\bar{x}, s$	Fact about sample	Average marks of the 500 surveyed

PRO TIP — The Key Distinction

If the number describes the entire population it is a parameter. If it describes a sample it is a statistic. India's decennial census produces parameters. An election exit poll produces statistics — with a margin of error.

03 Why Businesses Use Statistics

Revenue figures alone cannot tell a manager whether performance was good or bad. Statistics provides context — mean, variance, trend — to turn raw numbers into decisions. Companies that embed statistical thinking consistently outperform intuition-driven peers across industries.

DATA-DRIVEN COMPANIES OUTPERFORM PEERS BY

5–6×

Source: McKinsey Global Institute, *The Age of Analytics*, 2016

CASE STUDY — DMart (Avenue Supermarts), Ahmedabad

Context — DMart operates 100+ stores across Gujarat. Ahmedabad is one of its highest-volume markets with stores in Satellite, Maninagar, Bopal, and Chandkheda.

Challenge — With thousands of SKUs per store, the buying team needed to identify consistently underperforming products before overstock drained working capital.

What they did — Analysts computed mean weekly sales and standard deviation per SKU per store. Products with mean below threshold and high variance were flagged weekly for review.

Result — Underperforming SKU count reduced 12% in one quarter; faster-moving items received better shelf allocation and higher reorder quantities.

Takeaway — Basic descriptive statistics — mean and spread — are sufficient to drive significant operational improvement at scale.

Source: Avenue Supermarts Annual Report 2022–23 ([avenuessupermarts.com](https://www.avenuessupermarts.com))

04 Demo — Sample vs Population in Excel

Open the class dataset of 1,000 customer transactions. Compute `=AVERAGE()` across all 1,000 rows — this is the population mean. Then compute `=AVERAGE()` on only the first 100 rows — this is a sample statistic. The gap between the two values is sampling error, and it shrinks as sample size grows.

TRY THIS LIVE

In Excel: `=AVERAGE(B2:B1001)` for full population mean. `=AVERAGE(B2:B101)` for a sample of 100. `=AVERAGE(B2:B26)` for a sample of 25. Observe how the gap between sample and population mean widens as sample size shrinks — this is sampling error in action.

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 1, Lecture 1: Introduction to Statistics (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

Always link the official channel — beware fan-made or re-upload copies.

WHERE TO GO NEXT

You know the two branches of statistics and why they exist — every test and model in this module builds on this foundation.

Explore deeper — NCERT Class 11 Mathematics, Chapter 15: Statistics — free at ncert.nic.in. Short, clear, with Indian examples.

Try on your own — Find any published survey or news poll. Identify the sample size, the population being inferred, and whether the sample is likely representative.

Connect the dots — Session 2 introduces data types and measurement scales — the classification every statistical method depends on before any calculation begins.

Going further — Chapter 1 of 'The Signal and the Noise' by Nate Silver explores why even well-designed statistical models often fail in practice.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a statistics professor. Explain the difference between descriptive and inferential statistics using a Flipkart or Meesho example from India. Use simple Hinglish. Keep it under 150 words.

SESSION

02

60 MIN · FUNDAMENTALS

Types of Data and Measurement Scales

◆ Learning Objectives

- Classify data as numerical or categorical and identify subcategories of each
- Recognise continuous, discrete, nominal, ordinal, interval, and ratio data
- Explain why measurement scale determines which statistical method is valid
- Identify column data types from a real business dataset
- Apply correct data type recognition before any analysis step

01 Why Data Types Matter Before Any Analysis

Before any calculation, an analyst must know what kind of data they are working with. Applying a mean to a categorical variable — like averaging customer cities — is meaningless. Data type determines which statistics are valid and which will produce nonsense results.

CONCEPT — Two Major Data Categories

Numerical data holds measurable quantities — salaries, temperatures, product quantities. Categorical data represents labels or groups — gender, city, product category. Each category has subcategories, and each behaves differently in analysis.

02 Numerical Data — Continuous vs Discrete

Continuous

Can take infinite values in a range. Examples: Rs 47,382.50 revenue, 37.4 degC temperature, 2.73 km delivery distance. Use: mean, std deviation, histogram.

Discrete

Takes only countable whole values. Examples: 3 support tickets, 12 items ordered, 47 branches. Use: count, mode, bar chart.

Continuous data can take any value within a range — height, revenue, delivery time. Discrete data can only take whole, countable values — number of orders, number of complaints, headcount. The distinction determines whether you use a histogram (continuous) or a bar chart (discrete).

03 Categorical Data — Nominal vs Ordinal

Nominal data represents categories with no natural order — city names, gender, product colour. Ordinal data has a meaningful order but unequal spacing — customer satisfaction (1–5 stars), education level (SSC, HSC, Graduate). You can sort ordinal data but cannot do arithmetic on it meaningfully.

Type	Order?	Equal Spacing?	Example	Valid Statistics
Nominal	No	No	City, Gender, Product Colour	Mode, Count, Chi-square
Ordinal	Yes	No	Satisfaction rating, Education level	Median, Mode, Rank correlation
Interval	Yes	Yes	Year, Temperature in degC	Mean, SD (no true zero)
Ratio	Yes	Yes	Revenue, Height, Weight, Age	All statistics valid — true zero exists

04 Measurement Scales — Interval vs Ratio

Interval and ratio scales both have equal spacing, but the difference is the zero point. Ratio scale has a true zero — zero revenue means no revenue. Interval scale does not — year 0 is a calendar convention. This means ratios only make sense on ratio data: Rs 2 lakh is twice Rs 1 lakh; the year 2024 is not twice the year 1012.

PRO TIP — Quick Test for Scale

Ask: Does doubling the value mean twice as much? If yes -> Ratio scale (revenue, weight, age). If no -> Interval scale (year, temperature in degC). Ask: Does the order of categories matter? If no -> Nominal. If yes but spacing is unequal -> Ordinal.

TRY THIS LIVE

Open the Telco Churn dataset (Telco-Customer-Churn.csv). For each column, classify it: Nominal / Ordinal / Continuous / Discrete. Check: CustomerID, Gender, SeniorCitizen, Tenure, MonthlyCharges, TotalCharges, Churn. Which columns need encoding before modelling?

CASE STUDY — HDFC Bank Credit Analytics Team

Context — HDFC Bank processes millions of loan applications annually. Analysts work with a mix of data types: income (continuous), number of EMIs (discrete), education level (ordinal), city (nominal).

Challenge — A junior analyst applied an average to 'loan purpose' — a nominal variable — and got a

meaningless number. The team needed a clear data type framework.

What they did — The analytics team built a data dictionary for every variable, specifying type and valid statistical operations. Ordinal variables were encoded carefully before any model training.

Result — Model build time reduced because analysts no longer wasted effort on invalid operations. Feature engineering became systematic, improving model accuracy by an estimated 8%.

Takeaway — Knowing data types before analysis is not optional — it is the first gate every professional analyst must pass.

Source: HDFC Bank Annual Report 2022–23; IDRBT Working Paper on Bank Analytics

ANALYTICS PROJECTS FAILING DUE TO DATA QUALITY ISSUES

60%

Source: Gartner Data Quality Market Survey, 2021

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 1, Lecture 2: Types of Data and Scales of Measurement (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You can now classify any column in a dataset — this skill underpins every statistical method you will learn from Session 3 onwards.

Explore deeper — Chapter 1 of 'Statistics' by Freedman, Pisani and Purves — covers classification of data with examples. Available in most college libraries.

Try on your own — Take any dataset you have (Kaggle, company data, any CSV). List every column, its type, and which statistics are valid for it. Build this as a habit before every project.

Connect the dots — Session 3 starts computing — mean, median, mode — and the data types you identified here will determine when each measure is valid.

Going further — Read about Stevens' Theory of Measurement Scales (1946) — the original paper defining these four levels. Search 'Stevens 1946 measurement scales' on Google Scholar.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a data analyst. I will give you column names from a dataset. For each, tell me the data type (Nominal, Ordinal, Continuous, Discrete, Interval, or Ratio) and which statistical operations are valid. Columns:

CustomerID, City, Monthly Income, Number of EMIs, Satisfaction Score 1-5, Loan Status Approved or Rejected. Use an Indian banking context. Keep it under 150 words.

SESSION

03

60 MIN · DESCRIPTIVE STATISTICS

Measures of Central Tendency — Mean, Median, Mode

◆ Learning Objectives

- Calculate mean, median, and mode for a dataset manually and in Python
- Explain which measure of central tendency is appropriate in different contexts
- Demonstrate the effect of outliers on mean vs median
- Interpret central tendency results in a business context
- Apply the correct measure to a real sales dataset

01 Why Central Tendency?

Central tendency tells you where the centre of your data sits. If a branch manager asks 'What is a typical monthly sale?', they want one number that represents the whole dataset. Mean, median, and mode are three different answers to that question — each valid in different situations, and each misleading in the wrong one.

02 Mean — The Arithmetic Average

The mean is the sum of all values divided by the count. It works best when data is roughly symmetric without extreme outliers. The mean monthly revenue across 12 months gives a useful annual benchmark — as long as no single month was wildly different from the rest.

CONCEPT — When Mean Misleads

If Mukesh Ambani joins a room of 50 middle-income families, the mean income shoots up dramatically — but no longer represents anyone in the room. This is the Ambani effect. Whenever data has extreme outliers (top salaries, luxury purchases, exceptional months), mean misleads. Use median instead.

03 Median and Mode

The median is the middle value when data is sorted. It is resistant to outliers and is preferred for skewed distributions — this is why median household income is always reported, not mean. The mode is the value that appears most often — the only central tendency measure that works on categorical data.

Measure	How Calculated	Best For	Avoid When
Mean	Sum / Count	Symmetric data, no outliers	Skewed data, outliers present

Measure	How Calculated	Best For	Avoid When
Median	Middle value (sorted)	Skewed data, income, house prices	When frequency (mode) is needed
Mode	Most frequent value	Categorical data, survey responses	Continuous data with no repeats

PRO TIP — Quick Skewness Check

If Mean > Median -> data is right-skewed (a few very large values pulling mean up). If Mean < Median -> data is left-skewed. If Mean = Median -> roughly symmetric. This quick check guides your choice of measure before any formal test.

```
import numpy as np          Import NumPy
from scipy import stats     Import SciPy for mode
sales = [42000, 45000, 47000, 48000, 51000, 52000, 98000]

Monthly sales in Rs — last value is outlier
print('Mean: ', np.mean(sales))      54714 — pulled up by outlier
print('Median:', np.median(sales))    48000 — more representative
print('Mode: ', stats.mode(sales).mode[0])  Most frequent value
```

TRY THIS LIVE

In Excel: =AVERAGE() for mean, =MEDIAN() for median, =MODE() for mode on the sales dataset. Then manually add one extreme value such as Rs 5,00,000 in the last row. Recalculate all three. Observe how much the mean moves vs the median. This is the outlier effect.

04 Weighted Mean and Business Applications

Sometimes not all values contribute equally. A weighted mean assigns each value a weight based on its importance. Example: computing overall pass rate across branches of different sizes — a branch with 200 students should count more than one with 20 students. In Excel: =SUMPRODUCT(values, weights) / SUM(weights).

CASE STUDY — Amul Dairy, Anand, Gujarat

Context — Amul, headquartered in Anand, Gujarat, is Asia's largest dairy co-operative, with over 36 lakh farmer members contributing milk daily.

Challenge — Amul needed to set a fair procurement price for farmer members. Using the mean milk price was distorting the baseline because large farms with high volumes skewed the average upward.

What they did — The pricing team switched to median procurement price per fat percentage, segmented by

district. Mode was used to identify the most common supply quantity per district for fleet planning.

Result — Farmer satisfaction improved as prices better reflected the typical farmer's reality. Fleet utilisation improved 9% due to accurate mode-based volume planning.

Takeaway — Mean, median, and mode each serve different business questions — choosing the right one is as important as knowing how to calculate it.

Source: Amul Annual Report 2022–23 (amul.com)

AMUL DAILY MILK PROCUREMENT

3.5 Cr L

Source: GCMMF (Amul) Annual Report 2022–23

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 1, Lecture 3: Measures of Central Tendency (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You can now summarise the centre of any dataset — but the centre alone is incomplete. Session 4 covers spread and dispersion.

Explore deeper — Khan Academy — Mean, median and mode — free, with practice problems at khanacademy.org.

Try on your own — Take any 20-row dataset. Calculate mean, median, and mode of the key numeric column. Which best represents typical for this data? Explain your reasoning.

Connect the dots — Session 4 covers variance and standard deviation — how spread out data is around the mean. Together, central tendency plus dispersion form the core of descriptive statistics.

Going further — Look up geometric mean and harmonic mean — used in finance (CAGR calculations) and averaging rates. When would you use them instead of arithmetic mean?

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a data analyst at a retail company. I have monthly sales data: 42000, 45000, 47000, 48000, 51000, 52000, 98000 rupees. Calculate and interpret the mean, median, and mode. Tell me which best represents typical sales and why. Use simple Hinglish. Keep it under 150 words.

SESSION

04

60 MIN · DESCRIPTIVE STATISTICS

Measures of Dispersion — Range, Variance, Standard Deviation

◆ Learning Objectives

- Calculate range, variance, and standard deviation for a given dataset
- Interpret standard deviation in a business context
- Explain the coefficient of variation and when to use it for comparison
- Demonstrate dispersion calculations in Python
- Identify when high vs low dispersion signals a business problem or opportunity

01 Why Dispersion Matters

Two TOPS branches both average Rs 5 lakh monthly revenue. Branch A ranges from Rs 4.8L to Rs 5.2L every month — stable and predictable. Branch B ranges from Rs 1L to Rs 9L — chaotic and unpredictable. The mean hides this difference entirely. Dispersion measures reveal it.

CONCEPT — Dispersion Defined

Dispersion measures how spread out data values are from the centre. Low dispersion means values cluster tightly around the mean — predictable, consistent. High dispersion means values are scattered widely — volatile, unpredictable. Both scenarios have direct business implications.

02 Range — Simplest Measure of Spread

Range = Maximum - Minimum. It is the simplest dispersion measure but the least robust — it depends entirely on two extreme values and ignores everything in between. A single outlier dramatically inflates the range. Useful for a quick first look but never rely on it alone for business decisions.

03 Variance and Standard Deviation

Variance measures the average of squared deviations from the mean. Standard deviation (SD) is the square root of variance, expressed in the same unit as the original data. A low SD means values cluster close to the mean. A high SD means they are spread widely. SD is the most widely used dispersion measure in analytics.

CONCEPT — Population vs Sample SD

Population variance divides by N. Sample variance divides by (N-1) — called Bessel's correction — to compensate for the fact that a sample tends to underestimate population spread. Always use sample SD in analytics unless you have the entire population. In Python: `np.std(data, ddof=1)`. In Excel: `=STDEV.S()`.

```
import numpy as np                                Import NumPy
spend = [1200,1450,1380,1500,1100,1600,2800]      Customer spend in Rs — last is outlier
print('Range: ', np.max(spend) - np.min(spend))
Range: 1700
print('Variance:', round(np.var(spend, ddof=1)))
Sample variance — Bessel corrected
print('Std Dev: ', round(np.std(spend, ddof=1)))
SD in same unit as data
cv = np.std(spend, ddof=1)/np.mean(spend)*100     Coefficient of Variation
print(f'CV: {cv:.1f}%')                          Unit-free — enables fair comparison
```

04 Coefficient of Variation — Comparing Across Different Scales

Standard deviation cannot be compared across datasets with different units or means. The Coefficient of Variation (CV = SD / Mean × 100%) is unit-free and allows fair comparison. A network latency with CV of 12% is far less volatile than customer spend with CV of 45%, even if their absolute SDs look similar.

Measure	Formula	Unit	Strength	Weakness
Range	Max - Min	Same as data	Fast, simple	Sensitive to outliers
Variance	Avg squared deviations	Squared units (e.g. Rs ^2)	Mathematically tractable	Not intuitive
Std Deviation	sqrt(Variance)	Same as data	Most widely used, intuitive	Sensitive to outliers
CV	SD / Mean x 100%	% — unit-free	Enables cross-variable comparison	Not valid when mean is near zero

PRO TIP — The 68-95-99.7 Rule

For normally distributed data: 68% of values lie within 1 SD of the mean; 95% within 2 SD; 99.7% within 3 SD. This Empirical Rule lets you quickly spot outliers — any value beyond 3 SD is highly unusual and worth investigating.

TRY THIS LIVE

In Excel: =VAR.S() for sample variance, =STDEV.S() for sample standard deviation. Compute SD for the monthly revenue of two TOPS branches. Calculate CV = STDEV.S / AVERAGE for both. Which branch is more consistent? Discuss with the class.

CASE STUDY — Jio Network Quality Analytics, Ahmedabad

Context — Reliance Jio is India's largest telecom operator. Its network operations team tracks call drop rates, latency, and data speed across thousands of towers in Gujarat.

Challenge — Two towers both showed an average download speed of 42 Mbps. Customer complaints were far higher for one tower. The mean alone could not explain the difference.

What they did — The analytics team computed SD and CV for hourly download speed per tower. Tower A had SD of 2 Mbps — CV 4.8%, very consistent. Tower B had SD of 18 Mbps — CV 42.9%, highly variable during peak hours.

Result — Tower B was flagged for hardware upgrade. Within 60 days, its CV dropped to 8% and customer complaints from that zone reduced 67%.

Takeaway — Dispersion metrics reveal problems that averages hide. Two equal means can represent vastly different customer experiences.

Source: Reliance Jio Infocomm Annual Report 2022–23; TRAI Network Performance Report

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 2, Lecture 4: Measures of Dispersion — Variance and Standard Deviation (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You can now quantify both the centre and the spread of data — the two pillars of descriptive statistics.

Explore deeper — 'Naked Statistics' by Charles Wheelan — Chapter 2 on variance and what it hides. Practical and entertaining read.

Try on your own — Take any 20-day sales dataset. Calculate mean, SD, and CV. Identify days that fall outside 2 SD from the mean. What could explain those outlier days?

Connect the dots — Session 5 extends this into percentiles and boxplots — a powerful visual tool for comparing dispersion across groups.

Going further — Research 'robust measures of dispersion' — MAD (Mean Absolute Deviation) and IQR are alternatives to SD that are resistant to outliers. When would you prefer them?

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a data analyst. Explain variance and standard deviation to a non-technical business manager. Use an example of comparing monthly sales consistency across two TOPS Technologies branches in Ahmedabad. Use simple Hinglish. Keep it under 150 words.

SESSION

05

60 MIN · DESCRIPTIVE STATISTICS

Percentiles, Quartiles and Boxplots

◆ Learning Objectives

- Define percentile and calculate any Nth percentile for a dataset
- Compute Q1, Q2, Q3, and IQR from a dataset
- Identify outliers using the 1.5×IQR rule
- Interpret a boxplot and extract business insights from it
- Create a boxplot using Python and explain it to a non-technical audience

01 What Are Percentiles?

The Nth percentile is the value below which N% of observations fall. If a student's score is at the 85th percentile in a Gujarat state board exam, 85% of students scored below them. Percentiles allow ranking and comparison without knowing the full dataset — essential in salary benchmarking, exam results, and customer segmentation.

CONCEPT — Special Percentiles You Must Know

The 25th percentile is Q1 (first quartile), 50th is Q2 (median), and 75th is Q3 (third quartile). These three divide sorted data into four equal groups. The Interquartile Range ($IQR = Q3 - Q1$) covers the middle 50% of data — the most stable, representative portion, and is resistant to outliers.

02 Quartiles, IQR, and Outlier Detection

IQR is a robust measure of spread that ignores the top 25% and bottom 25% of values. Any value below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ is flagged as an outlier. This is not a rigid rule — it is a signal for investigation. A 3-hour delivery time may be a data error, an exceptional event, or a genuine operational failure.

Measure	Definition	Business Use
Q1 (25th pctl)	25% of data falls below this	Low-end customer segment boundary
Q2 / Median (50th)	50% below, 50% above	Typical customer value
Q3 (75th pctl)	75% of data falls below this	High-value customer threshold
IQR ($Q3 - Q1$)	Spread of middle 50%	Operational consistency measure
Outlier boundary	$< Q1 - 1.5 \times IQR$ or $> Q3 + 1.5 \times IQR$	Flag for investigation

PRO TIP — Outlier Does Not Mean Error

A statistical outlier is just an unusual value. It may be a data entry mistake (delete it), a legitimate exceptional event (keep it, explain it), or a fraud signal (escalate it). Always investigate the business context before removing any outlier from your dataset.

```
import numpy as np
import matplotlib.pyplot as plt
times = [18, 22, 25, 27, 28, 29, 30, 31, 32, 35, 38, 42, 90]

Delivery times in minutes — 90 is outlier
Q1 = np.percentile(times, 25)           First quartile
Q3 = np.percentile(times, 75)           Third quartile
IQR = Q3 - Q1
upper = Q3 + 1.5 * IQR                  Outlier boundary
print(f'Q1={Q1}, Q3={Q3}, IQR={IQR}, Upper bound={upper:.1f}')
plt.boxplot(times, vert=False)           Horizontal boxplot
plt.title('Delivery Times'); plt.show()
```

03 Reading a Boxplot

A boxplot visualises five statistics: minimum (excluding outliers), Q1, median (Q2), Q3, and maximum (excluding outliers). Outliers appear as individual dots beyond the whiskers. A long upper whisker means right-skewed data. A centred median line means roughly symmetric data. Side-by-side boxplots let you compare groups instantly.

TRY THIS LIVE

In Python: take the case study delivery time dataset. Run `plt.boxplot()` on two groups side by side — weekend vs weekday deliveries. Which group has higher IQR? Which has more outliers? Write one business insight from comparing the two boxes.

CASE STUDY — Zomato Delivery Analytics, Ahmedabad

Context — Zomato operates one of its highest-density delivery networks in Ahmedabad, active across CG Road, SG Highway, and Maninagar zones.

Challenge — Customer complaints about inconsistent delivery times were rising despite the average time being within the SLA of 35 minutes. Management needed to understand spread, not just the average.

What they did — The analytics team computed Q1, Q3, and IQR by zone and time slot. During 12–2 PM and 7–9 PM, IQR spiked to 28 minutes — meaning the middle 50% of deliveries varied by nearly half an hour.

Result — Peak-hour capacity planning was revised for 3 zones. IQR during peak slots reduced from 28 to 14 minutes within 90 days. Customer NPS improved 11 points.

Takeaway — IQR and boxplots reveal operational consistency problems that averages hide. Customers experience variance, not averages.

Source: Zomato Annual Report 2022–23 (zomato.com/investor-relations)

ZOMATO ORDERS PER DAY INDIA-WIDE

30+ Lakh

Source: Zomato Annual Report FY 2022–23

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 2, Lecture 5: Percentiles, Quartiles and Boxplots (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You can now measure, visualise, and interpret data spread using IQR and boxplots — a core EDA skill every analyst uses daily.

Explore deeper — Seaborn docs on `sns.boxplot()` — try the examples with the built-in tips dataset at seaborn.pydata.org.

Try on your own — On any dataset, create a grouped boxplot comparing two categories such as revenue by region. Write 3 business observations from the plot.

Connect the dots — Session 6 begins Probability — the mathematical foundation of all inferential statistics from Session 12 onwards.

Going further — Research violin plots — they add a kernel density estimate to the boxplot, giving more information about the full distribution shape, not just five summary statistics.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a data analyst presenting to a non-technical manager. Explain what a boxplot shows using an Ahmedabad food delivery company example. What do Q1, Q3, IQR, and the outlier dots mean in plain language? Use simple Hinglish. Keep it under 150 words.

SESSION

06

60 MIN · PROBABILITY

Probability Basics

Learning Objectives

- Define probability and distinguish between sample space, events, and outcomes
- Calculate classical and empirical probability for simple and compound events
- Apply the addition rule for mutually exclusive and non-exclusive events
- Distinguish between independent and mutually exclusive events
- Compute the probability of a customer purchasing a product using observed data

01 What is Probability?

Probability is a measure of how likely an event is to occur, expressed as a number between 0 and 1. A probability of 0 means impossible; 1 means certain. Every prediction in data analytics — from customer churn to loan default — is ultimately a probability estimate. Understanding probability is the gateway to all inferential statistics.

CONCEPT — Core Probability Terms

An experiment is any process that produces an outcome (a customer visiting your website, a transaction being made). The sample space is the set of all possible outcomes. An event is a subset of the sample space — the specific outcomes you are interested in. Classical probability = Favourable outcomes / Total outcomes.

02 Classical vs Empirical Probability

In a dataset of 10,000 Ahmedabad customers, if 2,400 purchased Product A, the probability of a random customer having purchased it is $2,400 / 10,000 = 0.24$, or 24%. This is empirical (observed) probability — based on historical data rather than theory — and it is the type analysts use most often in practice.

Type	Definition	Example	Formula
Classical	Based on equally likely outcomes	Probability of drawing an ace from a deck	Favourable / Total outcomes
Empirical	Based on observed data	P(customer churns) from past records	Event frequency / Total observations
Subjective	Based on expert judgment	P(new product succeeds)	Expert estimate — no fixed formula

03 Complement, Union, and Intersection

The complement of event A is the probability that A does NOT occur: $P(A') = 1 - P(A)$. The union is the probability that A OR B (or both) occurs. The intersection is the probability that A AND B both occur. These three rules are the building blocks for all compound probability calculations in analytics.

CONCEPT — Addition Rule

For mutually exclusive events (cannot both happen at once): $P(A \text{ or } B) = P(A) + P(B)$. For non-exclusive events (can both happen): $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. The subtraction removes the double-counted overlap.

```
total = 10000                                Total customers
bought_A = 2400; bought_B = 3100; bought_both = 800
Event counts
P_A = bought_A / total                       P(A) = 0.24
P_B = bought_B / total                       P(B) = 0.31
P_both = bought_both / total                 P(A and B) = 0.08
P_AorB = P_A + P_B - P_both                  Addition rule: P(A or B) = 0.47
print(f'P(A or B) = {P_AorB}')
```

Output: 0.47

04 Independent vs Mutually Exclusive Events

Mutually Exclusive

Cannot both happen. $P(A \text{ and } B) = 0$. Example: a customer being on both the Free and Premium tier simultaneously. $P(A \text{ or } B) = P(A) + P(B)$.

Independent

Occurrence of one does not affect the other. $P(A \text{ and } B) = P(A) \times P(B)$. Example: a customer buying electronics and buying clothing in separate visits.

TRY THIS LIVE

Using a customer dataset: compute $P(\text{customer is from Ahmedabad})$, $P(\text{customer purchased in December})$, $P(\text{from Ahmedabad AND purchased in December})$. Then compute $P(\text{from Ahmedabad OR purchased in December})$ using the addition rule. Verify your answer using Excel COUNTIFS().

CASE STUDY — ICICI Lombard General Insurance, Gujarat

Context — ICICI Lombard is one of India's leading general insurers with a large customer base in Gujarat covering motor, health, and home insurance policies.

Challenge — The risk analytics team needed to estimate the probability that a Surat customer aged 25–35

who owns a two-wheeler would file a claim in the next 12 months.

What they did — Analysts used historical claim data to compute empirical probabilities for each risk factor. They combined these using multiplication for independent factors and the addition rule for overlapping risk categories.

Result — The model produced customer-level risk scores that improved premium pricing accuracy. Loss ratio improved by 4 percentage points in that segment within one year.

Takeaway — Probability is not abstract theory — it directly drives pricing, risk management, and profitability in insurance and finance.

Source: ICICI Lombard Annual Report 2022–23 (icicilombard.com)

PRO TIP — Empirical Probability is Your Best Friend

In analytics you almost never work with theoretical (classical) probability. You almost always compute empirical probability from data: count the occurrences, divide by total observations. This is the foundation of churn rates, conversion rates, default rates, and every KPI that ends in the word 'rate'.

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 3, Lecture 6: Introduction to Probability — Sample Space and Events (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You now understand how to quantify likelihood — the language of all predictions, models, and decisions under uncertainty.

Explore deeper — Khan Academy — Basic Probability unit at [khanacademy.org](https://www.khanacademy.org/math/basic-probability) — free, interactive, with visual explanations.

Try on your own — From any business dataset, select two events and calculate their individual probabilities, intersection, and union. Check whether the events are independent or mutually exclusive.

Connect the dots — Session 7 introduces conditional probability and Bayes Theorem — how prior knowledge updates probability estimates. This is the engine behind spam filters, recommendation systems, and churn models.

Going further — Read about the Frequentist vs Bayesian approaches to probability — two fundamentally different philosophies on what probability means. Search Bayesian vs frequentist statistics on Towards Data Science.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a probability tutor. Explain the addition rule for probability using an Indian e-commerce example such as customers who buy electronics OR clothing on Flipkart. Include the case of overlapping events. Use simple Hinglish. Keep it under 150 words.

SESSION

07

60 MIN · PROBABILITY

Conditional Probability and Bayes' Theorem

◆ Learning Objectives

- Define conditional probability and explain how it differs from joint probability
- Apply Bayes Theorem to update probabilities with new evidence
- Interpret prior, likelihood, and posterior in a business scenario
- Compute the probability of cross-purchase using conditional probability
- Recognise where Bayes Theorem is applied in real analytics products

01 Conditional Probability — $P(A|B)$

Conditional probability is the probability of event A occurring given that event B has already occurred. Written $P(A|B)$, it updates our estimate based on known information. If you know a customer has already bought a mobile phone, the probability they will also buy a case is higher than for the general population — this is conditional probability in action.

CONCEPT — The Formula

$P(A|B) = P(A \text{ and } B) \text{ divided by } P(B)$. Read as: the probability of A given B equals the joint probability of both occurring divided by the probability of B alone. This requires $P(B)$ to be greater than zero — you cannot condition on an event with zero probability.

02 Worked Example — Cross-Purchase Probability

In a dataset of 10,000 Flipkart customers: 3,000 bought a phone (event A), 1,500 bought accessories (event B), 900 bought both. $P(B|A) = (900/10,000) / (3,000/10,000) = 0.09 / 0.30 = 0.30$. Given a customer bought a phone, there is a 30% chance they also bought accessories — twice the base rate of 15%.

```
total = 10000                                Total customers
bought_phone = 3000; bought_acc = 1500; bought_both = 900
Event counts
P_B_given_A = bought_both / bought_phone      P(accessories | phone) = 0.30
P_B_base = bought_acc / total                  Base rate P(accessories) = 0.15
print('Base rate:', P_B_base)                  0.15
print('Given phone buyer:', P_B_given_A)      0.30 — twice the base rate
# Use this to target phone buyers with accessory promotions
Business insight
```

03 Bayes Theorem — Updating Beliefs with Evidence

Bayes Theorem allows us to reverse conditional probabilities — to calculate $P(A|B)$ when we know $P(B|A)$. It is the mathematical foundation for updating belief when new evidence arrives. Named after Reverend Thomas Bayes (18th century), it powers modern ML classifiers, spam filters, and medical diagnostics.

CONCEPT — Bayes Theorem

$P(A|B) = P(B|A)$ times $P(A)$, all divided by $P(B)$. Three components: Prior $P(A)$ — belief before new evidence. Likelihood $P(B|A)$ — probability of the evidence if A is true. Posterior $P(A|B)$ — updated belief after seeing evidence. The posterior becomes the new prior when the next piece of evidence arrives.

Prior Probability

Our initial belief before seeing new data. Example: $P(\text{fraud}) = 2\%$ based on historical fraud rate across all transactions. This is what we start with.

Posterior Probability

Updated belief after new evidence. Example: $P(\text{fraud} | \text{transaction from new device in foreign country}) = 35\%$ — much higher after conditioning on suspicious signals.

TRY THIS LIVE

Bayes problem: $P(\text{Disease}) = 1\%$ base rate. Test sensitivity $P(\text{Positive}|\text{Disease}) = 95\%$. $P(\text{Positive}|\text{No Disease}) = 10\%$. A patient tests positive. What is $P(\text{Disease}|\text{Positive})$? Apply Bayes Theorem. Hint: $P(\text{Positive}) = P(\text{Pos}|\text{Disease}) \times P(\text{Disease}) + P(\text{Pos}|\text{No Disease}) \times P(\text{No Disease})$. Surprising answer: less than 9%.

CASE STUDY — Flipkart Recommendation Engine

Context — Flipkart, India's largest e-commerce platform, serves over 450 million registered users. Its Bengaluru-based engineering team builds recommendation systems used by millions of Indian shoppers daily.

Challenge — Showing every customer the same recommendations (base rate approach) led to low click-through rates. The team needed personalisation at scale.

What they did — The recommendation system uses a Bayesian approach: given a customer's browsing history (evidence), the system updates the prior probability of purchase for each product category. Customers who viewed camera accessories had a high posterior probability for memory cards and tripods.

Result — Personalised Bayesian recommendations improved click-through rate by 35% and average order value by 12% in deployed segments.

Takeaway — Bayes Theorem is not just a formula — it is a framework for rational decision-making under uncertainty, embedded in products millions of Indians use daily.

Source: Flipkart Engineering Blog (tech.flipkart.com)

PRO TIP — Where Bayes Shows Up in Analytics

Naive Bayes classifier (text classification, spam detection), Bayesian A/B testing, medical diagnostics, credit scoring, fraud detection. Any time you update a prediction with new evidence — you are applying Bayesian thinking, even if you do not call it that.

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 3, Lecture 7: Conditional Probability and Bayes Theorem (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You can now update probability estimates with evidence — one of the most powerful mental models in all of analytics.

Explore deeper — 3Blue1Brown — Bayes theorem, the geometry of changing beliefs on YouTube. One of the best visual explanations available.

Try on your own — Find any binary classification scenario (fraud/not fraud, churn/retain). Define prior, likelihood, and compute a posterior using Bayes Theorem. Document your assumptions clearly.

Connect the dots — Session 8 covers discrete probability distributions — building on probability to describe entire patterns of outcomes rather than single events.

Going further — Research the Monty Hall Problem — a famous counterintuitive result from conditional probability. Search Monty Hall conditional probability explained for multiple solutions.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a statistics tutor. Explain Bayes Theorem using a practical Indian example — like testing for a disease in a population or fraud detection for an Indian bank. Walk through prior, likelihood, and posterior step by step. Use simple Hinglish. Keep it under 150 words.

SESSION

08

60 MIN · PROBABILITY

Probability Distributions — Discrete

◆ Learning Objectives

- Define a probability distribution and explain its role in analytics
- Apply the Bernoulli and Binomial distributions to binary outcome problems
- Use the Poisson distribution to model event counts per unit time
- Calculate probabilities using `scipy.stats` in Python
- Identify which discrete distribution fits a given business scenario

01 What is a Probability Distribution?

A probability distribution describes all possible outcomes of a random variable and the probability of each. Instead of calculating probability for a single event, a distribution gives you a complete map of probabilities across all possible events. This is the bridge from individual probabilities to predictive modelling and business forecasting.

CONCEPT — Random Variable

A random variable is a variable whose value is determined by a random process. Discrete random variables take countable values (0, 1, 2, 3) — number of daily orders, number of defects. Continuous random variables take any value in a range — weight, time, revenue. This session covers discrete distributions only.

02 Bernoulli and Binomial Distributions

The Bernoulli distribution models a single trial with exactly two outcomes: success (1) or failure (0). Parameter p = probability of success. The Binomial distribution extends this to n independent Bernoulli trials and counts the total successes. It answers: of 100 loan applications, how many are expected to default? Of 500 emails sent, how many will be opened?

```
from scipy.stats import binom
```

```
n, p = 100, 0.12
```

```
print(binom.pmf(10, n, p))
```

```
print(binom.cdf(15, n, p))
```

```
print('Mean:', n*p)
```

```
import numpy as np
```

```
print('SD: ', round(np.sqrt(n*p*(1-p)),2))
```

Binomial distribution

100 applications; 12% default rate

$P(\text{exactly 10 defaults})$ — probability mass

$P(15 \text{ or fewer defaults})$ — cumulative

Expected defaults: 12

Standard deviation of defaults

03 Poisson Distribution — Counting Events in Time

The Poisson distribution models the number of events occurring in a fixed time period when events occur independently at a constant average rate. It is ideal for: number of daily support tickets, orders per hour, server errors per day, customer arrivals per minute. The key parameter is lambda (lambda) — the average count per period.

CONCEPT — Poisson Properties

Mean = lambda. Variance = lambda — same as the mean, which is a unique property of the Poisson distribution. When lambda is large (above 30), the Poisson approximates a normal distribution. When lambda is small and events are rare, Poisson is the right choice — for example, number of fraud transactions per day.

```
from scipy.stats import poisson
lam = 8
print(poisson.pmf(10, lam))
print(poisson.cdf(12, lam))
print(1 - poisson.cdf(15, lam))
```

Poisson distribution
Average 8 support tickets per hour
P(exactly 10 tickets this hour)
P(12 or fewer tickets — manageable)
P(more than 15 — team overwhelmed)

Distribution	Key Scenario	Parameter(s)	Business Use
Bernoulli	Single yes/no trial	p = success probability	Will one customer churn?
Binomial	n trials, count successes	n, p	How many of 500 emails will convert?
Poisson	Count events in fixed time	lambda = average rate	How many tickets per hour? Orders per minute?

TRY THIS LIVE

Ola's Ahmedabad city hub receives an average of 15 ride requests per minute during peak hours. Use Poisson distribution (`scipy.stats.poisson`) to find: P(exactly 20 requests in one minute), P(fewer than 10 requests). What does this tell a fleet manager about capacity planning?

CASE STUDY — Ola Ride Demand Forecasting, Ahmedabad

Context — Ola Electric and Ola Cabs operate high-frequency ride services in Ahmedabad, with peak demand near GIFT City, SG Highway, and Navrangpura during morning and evening rush hours.

Challenge — The operations team needed to pre-position drivers before peak hours, not reactively after demand spiked. They needed to quantify the probability demand would exceed fleet capacity.

What they did — Analysts modelled ride requests per 5-minute interval using the Poisson distribution with lambda = average requests per interval by zone and time slot. They computed P(requests greater than fleet

capacity) for each zone.

Result — Driver pre-positioning reduced average wait time from 4.8 to 3.1 minutes in high-demand zones. Driver utilisation improved 18%.

Takeaway — Poisson distribution is the go-to model for any business question about event frequency — queues, requests, failures, and arrivals.

Source: Ola Annual Report 2022–23 (olacabs.com)

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 4, Lecture 8: Discrete Probability Distributions — Binomial and Poisson (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You can now model binary outcomes and event counts probabilistically — the basis of many predictive models in analytics.

Explore deeper — Think Stats by Allen Downey — free at greenteapress.com. Chapter on distributions includes Python code.

Try on your own — Find a business dataset with a binary outcome column (churn, conversion, default). Estimate p and compute the expected number of successes in 1000 trials using the Binomial distribution. Does it match the observed count?

Connect the dots — Session 9 introduces continuous distributions — particularly the normal distribution, which underpins all parametric tests from Session 12 onwards.

Going further — Research Geometric distribution and Negative Binomial distribution — useful for modelling how long until the first success, or overdispersed count data where variance exceeds mean.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a data science educator. Explain the Poisson distribution using an Indian example — like customer support ticket volume at a Bengaluru tech company or ride requests on Ola in Ahmedabad. When should I use Poisson instead of Binomial? Use simple Hinglish. Keep it under 150 words.

SESSION

09

60 MIN · PROBABILITY

Probability Distributions — Continuous

◆ Learning Objectives

- Explain the normal distribution and its key properties
- Calculate Z-scores to standardise values and compare across datasets
- Use Z-scores to determine probabilities from the standard normal curve
- Plot a normal distribution using Python and interpret its shape
- Identify whether a business dataset is approximately normally distributed

01 Continuous Distributions — Key Concepts

For continuous data, we cannot assign probability to a single exact point — the probability of a customer spending exactly Rs 47,382.50 is effectively zero. Instead we work with probability density functions (PDFs) that give probability over ranges. The area under a PDF between any two values gives the probability of falling in that range.

CONCEPT — Probability Density Function (PDF)

A PDF describes the relative likelihood of a continuous random variable taking a given value. The total area under the entire PDF equals 1. To find $P(a < X < b)$, compute the area under the curve between a and b . For the normal distribution, this is done using Z-scores or `scipy.stats.norm.cdf()`.

02 The Normal Distribution — The Most Important Distribution in Statistics

The normal distribution (bell curve) is symmetric around its mean, with 68% of values within 1 SD, 95% within 2 SD, and 99.7% within 3 SD (the Empirical Rule). Defined by mean (μ) and standard deviation (σ). Many real-world phenomena follow it approximately: heights, weights, exam scores, and measurement errors.

Range	% of Values	Business Interpretation
$\mu \pm 1\sigma$	68.27%	Most typical values — normal operations
$\mu \pm 2\sigma$	95.45%	Broad normal range — acceptable variation
$\mu \pm 3\sigma$	99.73%	Almost all values — beyond this is an outlier
Beyond $\pm 3\sigma$	0.27%	Rare events — investigate immediately

03 Z-Score — Standardising for Comparison

A Z-score measures how many standard deviations a value is from the mean: $Z = (X - \mu) / \sigma$. A Z-score of 0 means the value equals the mean. A Z-score of +2 means the value is 2 SDs above the mean. Z-scores allow comparison across different scales and units — essential for comparing datasets with different means and SDs.

CONCEPT — Standard Normal Distribution

When data is converted to Z-scores, the resulting distribution has mean = 0 and SD = 1. This is the Standard Normal Distribution, denoted $Z \sim N(0,1)$. `scipy.stats.norm.cdf()` gives the area under the curve to the left of any Z-score — i.e., the probability of being below that value.

```
import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt
mu, sigma = 45000, 8000
z = (60000 - mu) / sigma
print('Z-score:', round(z,3))
print('P(X>60000):', round(1 - norm.cdf(z),4))
    About 3.04%
x = np.linspace(mu-4*sigma, mu+4*sigma, 300)
plt.plot(x, norm.pdf(x, mu, sigma))
plt.title('Loan Amount Distribution'); plt.show()
```

Normal distribution

Mean loan amount Rs 45,000 — SD Rs 8,000

Z-score for Rs 60,000 = 1.875

1.875

Bell curve

TRY THIS LIVE

Using the customer transaction dataset: compute mean and SD of purchase amounts. Calculate the Z-score for a transaction of Rs 2,00,000. What is the probability of seeing a transaction this large or larger? Use `norm.cdf()` in Python. Is this a statistical outlier?

Normal Distribution

Symmetric, bell-shaped. Used for large samples, natural phenomena, averages. Parameters: mean and SD. Fully described by just these two values.

t-Distribution

Similar to normal but heavier tails — more probability in the extremes. Used when n is less than 30. Shape depends on degrees of freedom. Approaches normal as n increases.

CASE STUDY — HDFC Bank Loan Amount Outlier Detection

Context — HDFC Bank is India's largest private sector bank by assets, with major retail lending operations across Gujarat including home loans, personal loans, and vehicle finance.

Challenge — The risk team needed to identify loan applications unusual in size — either too small

(potentially mis-categorised) or too large (elevated default risk requiring enhanced review).

What they did — Analysts fit a normal distribution to historical loan amounts per product category. Z-scores were computed for every incoming application. Applications with absolute Z-score above 2.5 were auto-flagged for manual underwriting review.

Result — The system flagged 3.2% of applications as outliers. Of flagged cases, 28% resulted in modified loan terms or rejection — reducing the portfolio default rate by 1.8 percentage points.

Takeaway — Z-scores are a simple, powerful tool for outlier detection in continuous data — applicable wherever a normal distribution is a reasonable approximation.

Source: HDFC Bank Annual Report 2022–23 (hdfcbank.com)

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 4, Lecture 9: Normal Distribution and Z-Scores (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You can now work with the normal distribution — the distribution that underpins every parametric test in inferential statistics.

Explore deeper — StatKey at statkey.lock5stat.com — free browser tool to visualise normal distribution probabilities interactively.

Try on your own — Pick any numeric column from your case study dataset. Check if it is approximately normal: compute mean and median, plot a histogram. Does it look bell-shaped? Compare mean vs median.

Connect the dots — Session 10 introduces sampling techniques — how to draw valid samples, which determines the quality of all inferential conclusions.

Going further — Research log-normal distribution — income, house prices, and stock returns are often log-normally distributed (normal after taking the log). Why does this matter for analysis?

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a statistics educator. Explain the normal distribution and Z-score using an Indian example — like exam marks of Class 12 students in Gujarat or salary distribution at an IT company in Ahmedabad. What does a Z-score of +2 mean for a student? Use simple Hinglish. Keep it under 150 words.

SESSION

10

60 MIN · INFERENCE STATISTICS

Sampling Techniques

◆ Learning Objectives

- Explain why sampling is necessary in data analytics
- Describe and apply random, stratified, systematic, and cluster sampling
- Identify the appropriate sampling method for a given business scenario
- Discuss sample size and its effect on accuracy of estimates
- Generate a stratified sample from a customer dataset using Python

01 Why Sampling?

A national retail chain with 1 crore customers cannot survey all of them to understand satisfaction. A pharmaceutical company cannot give a new drug to the entire population to test it. Sampling allows us to draw conclusions about a population by studying a manageable, representative subset. The quality of conclusions depends entirely on the quality of the sample.

CONCEPT — Sampling Error vs Sampling Bias

Sampling error is the natural random difference between a sample statistic and the population parameter — it decreases as sample size increases. Sampling bias is a systematic error caused by a flawed sampling method — increasing sample size does not fix it. A biased sample gives wrong answers at any size.

02 Four Probability Sampling Methods

Method	How It Works	Best For	Risk
Simple Random	Every individual has equal chance	Homogeneous populations	May miss subgroups
Stratified	Divide into strata; sample each randomly	Heterogeneous populations (age, city, segment)	Requires knowing strata in advance
Systematic	Select every Kth item from a sorted list	Production quality control, ordered lists	Pattern in data can bias results
Cluster	Randomly select entire clusters	Geographically spread populations	Higher sampling error than stratified

03 Stratified Sampling — Most Used in Analytics

Stratified sampling is the most common method in business analytics because customer populations are rarely homogeneous. By dividing customers into meaningful strata (age bands, city tier, spend level) and sampling proportionally from each, we ensure the sample reflects the population structure and improve the precision of estimates.

```
import pandas as pd
df = pd.read_csv('customers.csv')          Load customer dataset
# Stratified sample – 10% from each city tier
sample = (df.groupby('city_tier', group_keys=False)
           .apply(lambda x: x.sample(frac=0.1, random_state=42)))
           Group by stratum
           Sample 10% from each group
print(sample['city_tier'].value_counts(normalize=True))
           Verify proportions match population
```

PRO TIP — Sample Size Rule of Thumb

For a margin of error of $\pm 5\%$ at 95% confidence, you need approximately 385 responses regardless of population size for very large populations. For $\pm 3\%$, you need about 1,068. The formula is $n = Z^2 \times p \times (1-p) / E^2$. Most business surveys use 385 to 1,000 respondents.

TRY THIS LIVE

Take the case study customer dataset. Identify 2–3 meaningful strata such as age group under 25, 25 to 40, and above 40. Check the proportion of each stratum in the full population. Use Python groupby apply sample to draw a stratified sample of 200 customers. Verify the proportions match the population.

CASE STUDY — Election Commission of India — Exit Poll Methodology

Context — India's general elections involve over 90 crore eligible voters across 543 parliamentary constituencies spanning wildly diverse demographic groups.

Challenge — Exit poll agencies (Axis My India, C-Voter, others) needed to predict results with limited time and resources. Survey design determines accuracy — a wrong method invalidates all results.

What they did — Leading agencies used stratified sampling by constituency with sub-stratification by urban/rural and demographic group. Sample sizes of 1,000–2,000 per constituency were set using margin-of-error calculations.

Result — Well-designed stratified polls consistently predicted overall results within 2–3 percentage points. Poorly designed quota samples in the same elections diverged by 10+ points.

Takeaway — Sample design is a professional skill. A wrong method with a large sample gives worse results

than a correct method with a smaller sample.

Source: CSDS-Lokniti Exit Poll Methodology Reports; Election Commission of India data

ELIGIBLE VOTERS IN INDIA — 2024 GENERAL ELECTION

96.8 Cr

Source: Election Commission of India, 2024

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 5, Lecture 10: Sampling Methods and Sample Size (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You now know how to design a sample that gives valid, generalisable results — a foundational skill for any analyst running surveys or A/B tests.

Explore deeper — Survey Methodology by Groves et al. — Chapter 4 on probability sampling. Available in most university libraries.

Try on your own — Design a sampling plan for this scenario: TOPS wants to survey 500 of its 1,00,000 past students about career satisfaction. What strata would you define? How many from each? What would your margin of error be?

Connect the dots — Session 11 covers the Central Limit Theorem — the mathematical guarantee that sample means converge to the population mean as sample size grows.

Going further — Research bootstrap sampling — a powerful computer-based technique that re-samples from observed data to estimate uncertainty without assuming any theoretical distribution.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a research analyst. Explain the difference between stratified and simple random sampling. Use an example of a satisfaction survey at a Gujarat IT company with employees from Engineering, Sales, HR, and Operations. Which method gives more accurate department-level insights? Use simple Hinglish. Keep it under 150 words.

SESSION

11

60 MIN · INFERENCE STATISTICS

Central Limit Theorem

◆ Learning Objectives

- State the Central Limit Theorem and explain its significance in statistics
- Demonstrate that the sampling distribution of the mean approaches normal as n increases
- Explain why the CLT enables all parametric hypothesis tests
- Simulate the CLT using Python with repeated samples
- Apply the CLT to calculate the probability of a sample mean exceeding a target

01 What is the Central Limit Theorem?

The Central Limit Theorem (CLT) states that the distribution of sample means approaches a normal distribution as sample size n increases, regardless of the shape of the original population distribution. This is arguably the most important theorem in all of statistics — it is the mathematical justification for why parametric tests work even when the population is not normally distributed.

CONCEPT — CLT in Three Lines

1. Take a population with any shape — skewed, uniform, bimodal, any shape at all. 2. Draw many samples of size n and calculate the mean of each sample. 3. As n grows (generally $n \geq 30$), the distribution of those sample means becomes approximately normal with mean = μ and SD = $\sigma/\sqrt{n}$, called the Standard Error.

02 Standard Error — The SD of Sample Means

The Standard Error ($SE = \sigma / \sqrt{n}$) is the standard deviation of the sampling distribution of the mean. It tells you how much sample means vary around the true population mean. A larger sample size produces a smaller SE — meaning sample means cluster more tightly around the true mean, giving more precise estimates.

Population SD (σ)

How much individual values vary within the population. Fixed property of the population. Example: SD of all Ahmedabad customer transactions = Rs 12,000.

Standard Error ($SE = \sigma/\sqrt{n}$)

How much sample means vary across different samples. Decreases as n grows. For $n=100$: $SE = Rs\ 12,000/100 = Rs\ 1,200$. For $n=400$: $SE = Rs\ 600$. Bigger sample, tighter estimate.

```
import numpy as np
import matplotlib.pyplot as plt
# Simulate CLT from a right-skewed (exponential) population
```

```
population = np.random.exponential(scale=50000, size=100000)
    Skewed population — not normal
sample_means = []
for _ in range(2000):
    s = np.random.choice(population, size=50)
    sample_means.append(np.mean(s))
plt.hist(sample_means, bins=50, edgecolor='white')
    Distribution of means is normal!
plt.title('CLT: Sample Means from Skewed Population'); plt.show()
```

TRY THIS LIVE

Run the simulation above with $n = 5$, then $n = 30$, then $n = 100$. Plot the histogram of sample means each time. Observe how the distribution transforms from skewed to approximately normal as n increases. This is the CLT in action — one of the most satisfying things to see in statistics.

03 Why CLT Matters for All Hypothesis Tests

The t-test, ANOVA, and regression tests in Sessions 13–17 all assume that the sampling distribution of the test statistic is approximately normal. The CLT is the mathematical justification for this assumption — as long as $n \geq 30$, we can safely use these tests even if the underlying population is far from normally distributed.

PRO TIP — When CLT Breaks Down

The CLT requires: (1) samples are independent of each other, (2) sample size is large enough — $n \geq 30$ is a guideline, but heavily skewed or heavy-tailed distributions may require $n > 100$. If your data is extremely skewed, increase your sample size before applying any parametric test.

CASE STUDY — SEBI Market Returns Analysis

Context — SEBI (Securities and Exchange Board of India) monitors daily returns across BSE and NSE. Daily stock returns are notoriously non-normal — they have fat tails and occasional extreme outliers.

Challenge — Regulatory analysts needed to use parametric risk models (based on normal distribution) to assess systemic risk, despite knowing individual stock returns are not normally distributed.

What they did — By aggregating returns across portfolios of 50+ securities and using rolling monthly averages, the CLT guarantees the portfolio-level mean returns are approximately normally distributed — enabling valid use of parametric risk models.

Result — VaR (Value at Risk) and stress testing models based on CLT-justified normality became the standard framework for SEBI-regulated risk reporting by mutual funds and banks.

Takeaway — CLT allows analysts to apply normal-distribution-based tools even in domains where individual data is clearly non-normal — as long as sample sizes are adequate.

Source: SEBI Annual Report 2022–23; RBI-SEBI joint working paper on systemic risk

NSE DAILY EQUITY TURNOVER

Rs 75,000 Cr+

Source: NSE India, FY 2022–23 average

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 5, Lecture 11: Central Limit Theorem and Sampling Distribution (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

The CLT is the mathematical bridge between descriptive and inferential statistics — you now have the foundation to understand every hypothesis test ahead.

Explore deeper — StatQuest — The Central Limit Theorem on YouTube by Josh Starmer. One of the clearest visual explanations available anywhere.

Try on your own — Take a clearly non-normal dataset (e.g., daily sales — often right-skewed). Draw 1,000 samples of $n=40$. Plot the sample means. Do they look approximately normal? How does this change for $n=5$?

Connect the dots — Session 12 introduces hypothesis testing — the formal framework for using sample statistics to make claims about populations. CLT is what makes it valid.

Going further — Research the Law of Large Numbers — related to but distinct from CLT. LLN guarantees that the sample mean converges to the population mean; CLT tells you the shape of that convergence process.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a statistics teacher. Explain the Central Limit Theorem in simple terms using an example of average daily sales at 30 TOPS branches in Gujarat. Why does the distribution of branch means look normal even if individual branch sales are skewed? Use Hinglish. Keep it under 150 words.

SESSION

12

60 MIN · INFERENCE STATISTICS

Hypothesis Testing Concepts

Learning Objectives

- Define null and alternative hypotheses and write them correctly for a business scenario
- Explain Type I and Type II errors and their business consequences
- Interpret p-values and confidence levels correctly
- Determine when to reject the null hypothesis at a given significance level
- Formulate a complete hypothesis test plan for a real business claim

01 What is Hypothesis Testing?

Hypothesis testing is a formal procedure to decide whether a claim about a population is supported by sample evidence. Instead of measuring the entire population, we use a sample to test whether the claim is likely true or false. It converts business questions — Did the new discount actually increase sales? — into statistical decisions backed by probability.

CONCEPT — The Two Hypotheses

The Null Hypothesis (H_0) is the default claim — usually that there is no effect, no difference, or no change. The Alternative Hypothesis (H_1) is the claim we want to prove — that there IS an effect. We never directly prove H_1 . We either reject H_0 (evidence for H_1) or fail to reject H_0 (insufficient evidence for H_1).

02 Type I and Type II Errors

Any hypothesis test can produce two types of errors. A Type I error (false positive) is rejecting H_0 when it is actually true — claiming an effect exists when it does not. A Type II error (false negative) is failing to reject H_0 when H_1 is actually true — missing a real effect. The consequences differ dramatically depending on the business domain.

Error Type	What Happened	Example	Consequence
Type I (alpha)	Rejected true H_0 — false positive	Concluding a drug works when it doesn't	Patient gets ineffective treatment
Type II (beta)	Failed to reject false H_0 — false negative	Missing a real increase in fraud rate	Fraud goes undetected
Significance Level (alpha)	P(Type I error) — set by analyst	Usually 0.05 (5%)	Lower alpha = stricter test, more Type II risk

Error Type	What Happened	Example	Consequence
Power (1-beta)	P(correctly detecting real effect)	Usually target $\geq 80\%$	Higher power = less likely to miss effects

CONCEPT — p-Value Explained

The p-value is the probability of observing your sample result (or more extreme) IF the null hypothesis were true. A small p-value (below 0.05) means the observed result is unlikely under H_0 — evidence against it. A large p-value means the data is consistent with H_0 . The p-value is NOT the probability that H_0 is true.

03 Decision Rules and Confidence Levels

The significance level α (usually 0.05) is the threshold for rejecting H_0 . If $p\text{-value} < \alpha \rightarrow$ reject H_0 . If $p\text{-value} \geq \alpha \rightarrow$ fail to reject H_0 . The confidence level (1- α) is 95% when $\alpha = 0.05$. A 95% confidence interval means: if we repeated this experiment 100 times, 95 of the resulting intervals would contain the true population parameter.

$p < 0.05$ — Reject H_0

The observed data is statistically unlikely under the null hypothesis. We have sufficient evidence to conclude H_1 . Result is statistically significant.
Example: new discount DID significantly increase sales.

$p \geq 0.05$ — Fail to Reject H_0

Insufficient evidence against H_0 . We do not prove H_0 is true — we simply lack evidence to reject it.
Example: the discount might have had no real effect; data cannot tell us either way.

PRO TIP — Statistical vs Practical Significance

A result can be statistically significant ($p < 0.05$) but practically meaningless. With $n = 1,000,000$, even a 0.001% improvement in conversion will be statistically significant — but is it worth acting on? Always report effect size (how large is the difference?) alongside the p-value.

```
# Hypothesis formulation example
# Claim: New discount increased average order value above Rs 450
# H_0: mu = 450 (no change)           Null hypothesis
# H_1: mu > 450 (one-tailed – we expect increase only)
#                                     Alternative hypothesis
alpha = 0.05                         Significance level: 5%
# Decision rule:
# If p-value < 0.05 -> reject H_0 -> discount had significant effect
# If p-value >= 0.05 -> fail to reject H_0 -> insufficient evidence
```

TRY THIS LIVE

Write H_0 and H_1 for each scenario: (1) A new training program claims to improve TOPS placement rate above the current 82%. (2) An A/B test checks whether a new website CTA increases sign-ups. (3) A factory claims its defect rate is under 2%. For each: one-tailed or two-tailed? What is a Type I error in business terms? What is a Type II error?

CASE STUDY — Marico Hypothesis Test for New Product Launch

Context — Marico, the FMCG company behind Parachute and Saffola, has manufacturing and R&D operations in Gujarat. It regularly tests product reformulations before a national rollout.

Challenge — Marico reformulated Saffola Active cooking oil. The product team claimed the new formula would improve repeat purchase rate from 34% to above 40% within 3 months. Marketing needed statistical evidence before committing Rs 15 crore to a national campaign.

What they did — H_0 : repeat purchase rate $\leq 34\%$. H_1 : repeat purchase rate $> 34\%$. A controlled test was run in 3 cities over 90 days with 800 panel households. Results tested at $\alpha = 0.05$.

Result — Sample repeat purchase rate: 41.2%. p-value: 0.003 — well below 0.05. H_0 rejected. National campaign approved. New formula achieved 42% repeat purchase rate nationally within 6 months.

Takeaway — Hypothesis testing provides the statistical confidence needed to back major business decisions with evidence rather than intuition.

Source: Marico Annual Report 2022–23 (marico.com); Nielsen FMCG panel data methodology

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 6, Lecture 12: Introduction to Hypothesis Testing — Null Hypothesis, p-values and Errors (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You now understand the hypothesis testing framework — every test in Sessions 13 through 17 is a direct application of exactly this logic.

Explore deeper — Statistics Done Wrong by Alex Reinhart — free at statisticsdonewrong.com. Chapter 3 on p-values and what they actually mean.

Try on your own — Take the case study dataset. Identify one testable business claim. Write H_0 and H_1 . Decide alpha and which type of error you most want to avoid. Justify your reasoning.

Connect the dots — Session 13 applies this framework to t-tests — the test for comparing means when you have one or two groups.

Going further — Search the replication crisis statistics — many published research findings have failed to

replicate. A large part of the cause is p-value misinterpretation and underpowered studies.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a statistics coach. Explain what a p-value means and what it does NOT mean, using a practical Indian business example — like an A/B test for a Flipkart product page. Correct the common misconception that $p < 0.05$ means there is a 95% probability the result is true. Use simple Hinglish. Keep it under 150 words.

SESSION

13

60 MIN · INFERENCE STATISTICS

t-Tests — One-Sample and Two-Sample

◆ Learning Objectives

- Explain when and why a t-test is used instead of a Z-test
- Conduct a one-sample t-test to compare a sample mean to a known standard
- Conduct an independent samples t-test to compare means of two groups
- Interpret t-statistic, p-value, and confidence interval from Python output
- Apply the appropriate t-test variant to a campaign performance comparison

01 When to Use a t-Test

A t-test is used to test whether a sample mean differs significantly from a target value (one-sample) or from another group's mean (two-sample). Use a t-test when: data is approximately normally distributed or $n \geq 30$ by CLT, population standard deviation is unknown (the usual case in practice), and you are comparing means of quantitative continuous data.

CONCEPT — t-Distribution vs Z-Distribution

The Z-test uses the normal distribution and requires knowledge of the population SD. In practice, population SD is almost never known — we use the sample SD instead. This introduces extra uncertainty, which the t-distribution accounts for through heavier tails. As sample size grows, the t-distribution approaches normal.

02 One-Sample t-Test

The one-sample t-test tests whether the mean of a sample is significantly different from a known or hypothesised population mean. Example: a branch claims average delivery time is 30 minutes. A sample of 25 deliveries gives a mean of 34 minutes. Is this difference statistically significant or just random variation?

```
from scipy.stats import ttest_1samp          One-sample t-test
delivery_times = [32, 35, 28, 36, 40, 31, 34, 38, 29, 33,
37, 35, 30, 34, 32, 36, 31, 38, 34, 40,
                29, 33, 35, 31, 36]
                25 delivery time samples in minutes
t_stat, p_value = ttest_1samp(delivery_times, popmean=30)
H_0: mean = 30 minutes
print(f't-statistic: {t_stat:.3f}')           How many SEs above the null mean
print(f'p-value:      {p_value:.4f}')         If < 0.05, reject H_0
```

03 Two-Sample Independent t-Test

The independent samples t-test compares the means of two unrelated groups. Example: does the marketing campaign in Ahmedabad generate higher average revenue per customer than the campaign in Surat? We collect samples from both cities and compare their means, accounting for within-group variation.

```
from scipy.stats import ttest_ind                Two-sample independent t-test
ahmedabad = [520, 480, 610, 590, 540, 505, 630, 498, 570, 525]
Revenue per customer, Ahmedabad (Rs 00s)
surat      = [460, 490, 440, 510, 470, 455, 480, 500, 462, 488]
Revenue per customer, Surat (Rs 00s)
t_stat, p_value = ttest_ind(ahmedabad, surat)    H_0: mu_1 = mu_2 (no difference)
print(f't-statistic: {t_stat:.3f}')
print(f'p-value:      {p_value:.4f}')            If < 0.05 -> significant difference
```

04 Paired t-Test — Before vs After

The paired t-test is used when the same subjects are measured twice — before and after an intervention. Since the same customers are measured, the pairs are related — this reduces variance and makes the test more sensitive to real changes. Example: comparing average revenue per customer before and after a marketing campaign in the same region.

Test	Groups	Subjects	Example
One-sample t-test	1 sample vs known mean	N/A	Is avg delivery = 30 min?
Independent t-test	2 separate groups	Different subjects	Ahmedabad vs Surat avg revenue
Paired t-test	1 group, 2 time points	Same subjects	Same customers before vs after campaign

PRO TIP — Welch's t-Test is the Safer Default

The standard independent t-test assumes equal variance in both groups (Levene's test checks this). Welch's t-test does not make this assumption — it works whether variances are equal or not. In scipy: `ttest_ind(a, b, equal_var=False)` uses Welch's. This is the recommended default for most real-world data.

TRY THIS LIVE

Using the case study revenue dataset: split data into before campaign and after campaign periods for the same region. Run `scipy.stats.ttest_rel()` for a paired t-test. Write: `H_0`, `H_1`, your alpha level, the t-statistic,

p-value, and your business conclusion. Was the campaign statistically significant?

CASE STUDY — Swiggy Delivery SLA Dispute, Ahmedabad

Context — Swiggy operates across all major Ahmedabad zones — Satellite, Bopal, Navrangpura, and Maninagar. Its SLA commits to delivery within 35 minutes for most restaurant partners.

Challenge — Customer complaints suggested actual delivery times were significantly above the 35-minute SLA, but the operations team disputed this. A statistical test was needed to settle the question.

What they did — A random sample of 200 deliveries was drawn. H_0 : mean delivery time = 35 minutes. H_1 : mean delivery time > 35 minutes (one-tailed). A one-sample t-test was run at $\alpha = 0.05$.

Result — Sample mean: 38.4 minutes. t-statistic: 3.12. p-value: 0.001. H_0 rejected. Operations invested in zone-level fleet rebalancing. Within 45 days, mean delivery time returned to 35.1 minutes.

Takeaway — A t-test transformed a dispute between teams into an evidence-based conversation — and drove an operational fix that measurably improved customer experience.

Source: Swiggy Engineering Blog (bytes.swiggy.com)

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 6, Lecture 13: t-Tests for One and Two Samples (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You can now test claims about one or two group means — one of the most used tests in analytics, A/B testing, and business research.

Explore deeper — Pingouin Python library at pingouin-stats.org — provides t-test output including effect size (Cohen's d) and statistical power in a single function call.

Try on your own — On any numeric column in your dataset, compare the means of two natural groups (weekday vs weekend, region A vs region B). Run the appropriate t-test and write a one-paragraph business conclusion.

Connect the dots — Session 14 introduces the Chi-Square test — hypothesis testing for categorical data, where t-tests cannot be applied.

Going further — Research Welch's t-test vs Student's t-test in depth. Understand when the assumption of equal variance is actually met, and what happens to Type I error rate when it is violated.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a data analyst. Explain the difference between a one-sample t-test, an independent two-sample t-test, and a paired t-test. Give an Indian business example for each. When should I use each one? Use simple Hinglish. Keep it under 150 words.

SESSION

14

60 MIN · INFERENCE STATISTICS

Chi-Square Test

Learning Objectives

- Explain when to use a Chi-square test instead of a t-test
- Construct a contingency table from raw categorical data
- Conduct a Chi-square test of independence using scipy.stats
- Interpret the chi-square statistic, p-value, and degrees of freedom
- Apply the chi-square test to a real business scenario with categorical variables

01 Why Chi-Square? — Testing Categorical Data

The t-test works for numeric means. But what if the question is: Does gender influence product preference? Is there a relationship between city tier and payment method? These involve categorical variables — and a t-test cannot be applied here. The Chi-square test is designed exactly for testing relationships between categorical variables.

CONCEPT — Two Uses of Chi-Square

Chi-square Goodness of Fit tests whether observed frequencies match an expected distribution (e.g., are customers equally distributed across four city tiers?). Chi-square Test of Independence tests whether two categorical variables are related or independent (e.g., is gender independent of product category purchased?).

02 Building a Contingency Table

The Chi-square test of independence starts with a contingency table: rows are categories of variable 1, columns are categories of variable 2, and cells contain observed frequencies. We then compute expected frequencies — what we would expect if the two variables were truly independent — and measure how much the observed diverges from the expected.

	Electronics	Clothing	Total
Male	120	80	200
Female	60	140	200
Total	180	220	400

Expected frequency for each cell = (Row Total × Column Total) / Grand Total. If observed frequencies are close to expected, the variables are independent. If they differ greatly, the variables are associated. The chi-square statistic measures the size of this difference across all cells simultaneously.

```
from scipy.stats import chi2_contingency
import numpy as np
observed = np.array([[120, 80],
                     [60, 140]])
chi2, p, dof, expected = chi2_contingency(observed)

Run the test
print(f'Chi2 statistic: {chi2:.3f}')
print(f'p-value: {p:.4f}')
print(f'Degrees of freedom: {dof}')
print('Expected:\n', expected.round(1))
```

Chi-square test

Male: 120 electronics, 80 clothing
Female: 60 electronics, 140 clothing

Larger = more deviation from independence
If < 0.05 -> significant association
(rows-1) $\times$ (cols-1)
What we expected under independence

03 Interpreting the Result

If $p\text{-value} < \alpha$ (0.05): reject H_0 — the two categorical variables are NOT independent; there IS a significant association. If $p\text{-value} \geq \alpha$: fail to reject H_0 — insufficient evidence of a relationship. The Chi-square test tells you WHETHER an association exists, not how strong it is. For strength, calculate Cramér's V.

PRO TIP — Chi-Square Assumptions

Chi-square requires: (1) observations are independent of each other, (2) expected frequency in EVERY cell ≥ 5 . If any expected cell count is below 5, use Fisher's Exact Test (for 2×2 tables) or merge categories. Violating this assumption inflates Type I error — you will see significance that is not real.

TRY THIS LIVE

Using the e-commerce dataset: create a contingency table of Payment Method (UPI, Card, COD) vs Customer Tier (Premium, Regular). Run `chi2_contingency()`. Is payment method choice independent of customer tier? Write H_0 , H_1 , and your business conclusion based on the p-value.

CASE STUDY — Reliance Retail — Gender vs Product Category, Gujarat

Context — Reliance Retail operates JioMart, Reliance Fresh, Smart Bazaar, and Trends stores across Gujarat. Category management teams need to understand whether product preferences differ by gender for store layout decisions.

Challenge — The buying team asserted that female customers prefer clothing and personal care while male customers prefer electronics. The analytics team needed statistical evidence to optimise floor layouts in Ahmedabad stores.

What they did — Transaction data from 3 Ahmedabad stores was aggregated. A contingency table of gender vs primary product category was constructed. Chi-square test of independence was run at $\alpha = 0.05$.

Result — Chi-square statistic = 42.7, $p\text{-value} < 0.001$. H_0 rejected. Significant association confirmed. Store

layouts were adjusted; cross-sell promotions redesigned. Basket size improved 7% in adjusted stores.

Takeaway — Chi-square tests uncover customer behaviour patterns in categorical data — directly informing merchandising and marketing decisions.

Source: Reliance Retail Annual Report 2022–23 (relianceretail.com)

RELIANCE RETAIL REGISTERED CUSTOMERS

24 Cr+

Source: Reliance Retail Annual Report FY 2022–23

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 7, Lecture 14: Chi-Square Test — Goodness of Fit and Independence (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You can now test relationships between categorical variables — essential for customer segmentation, market research, and survey analysis.

Explore deeper — Chi-square test explained on Towards Data Science — free article with Python examples and visual contingency tables.

Try on your own — On your churn dataset, test: Is churn status independent of contract type? Build the contingency table, run the test, and write a business recommendation based on the result.

Connect the dots — Session 15 introduces ANOVA — for comparing means across three or more groups simultaneously. ANOVA extends the t-test logic to multiple groups.

Going further — Research Cramér's V — the effect size measure for Chi-square. It ranges from 0 (no association) to 1 (perfect association) and is essential for reporting the strength, not just the existence, of categorical relationships.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a data analyst. Explain the Chi-square test of independence using an Indian retail example — like testing whether payment method (UPI, Cash, Card) is related to age group at a Reliance or DMart store. What is the null hypothesis? What does $p < 0.05$ mean in business terms here? Use simple Hinglish. Keep it under 150 words.

SESSION

15

60 MIN · INFERENCE STATISTICS

ANOVA — Analysis of Variance

◆ Learning Objectives

- Explain why ANOVA is needed when comparing more than two group means
- Define between-group and within-group variance and their ratio
- Conduct a one-way ANOVA using `scipy.stats.f_oneway` in Python
- Interpret the F-statistic and p-value from ANOVA output
- Run post-hoc Tukey HSD to identify which specific groups differ

01 Why ANOVA — The Multiple Groups Problem

If you have three groups (Ahmedabad, Surat, Vadodara customers) and want to compare mean purchase values, running three separate t-tests is wrong — each test has a 5% chance of a false positive, and running three tests inflates the overall Type I error to nearly 15%. ANOVA tests all groups simultaneously with a single test, keeping the error rate at 5%.

CONCEPT — What ANOVA Tests

ANOVA tests H_0 : all group means are equal ($\mu_1 = \mu_2 = \dots = \mu$) against H_1 : at least one group mean is different. It does this by comparing between-group variance (differences between group means) to within-group variance (natural variation within each group). The ratio is the F-statistic.

02 Between-Group vs Within-Group Variance

High F-statistic

Between-group variance is large relative to within-group variance. Group means are far apart relative to the scatter within groups. Result: p-value is small -> reject H_0 -> at least one group is significantly different.

Low F-statistic (near 1)

Between-group variance is small relative to within-group variance. Group means differ no more than natural noise. Result: p-value is large -> fail to reject H_0 -> no significant difference found.

```
from scipy.stats import f_oneway                                One-way ANOVA
under25 = [1200, 1350, 1100, 1400, 1250, 1180, 1320, 1290]
Age < 25 purchase values (Rs)
age25_40 = [2100, 2300, 1950, 2250, 2400, 2050, 2200, 2350]
Age 25–40
```

```
above40 = [3200, 3500, 2900, 3400, 3100, 3600, 2800, 3300]
```

Age > 40

```
F, p = f_oneway(under25, age25_40, above40)
```

Run ANOVA

```
print(f'F-statistic: {F:.3f}')
```

Large F -> means differ significantly

```
print(f'p-value: {p:.6f}')
```

If < 0.05 -> reject H₀

03 Post-Hoc Testing — Which Groups Differ?

A significant ANOVA result only tells you at least one group mean is different. It does NOT tell you which groups differ from which. For that, run a post-hoc test. Tukey's HSD (Honestly Significant Difference) is the most common — it compares all pairs of groups while controlling for the overall Type I error rate.

```
from statsmodels.stats.multicomp import pairwise_tukeyhsd
```

Post-hoc Tukey HSD

```
import numpy as np
```

```
values = np.concatenate([under25, age25_40, above40])
```

Combine all values

```
groups = ['<25']*8 + ['25-40']*8 + ['>40']*8
```

Group labels

```
result = pairwise_tukeyhsd(values, groups, alpha=0.05)
```

Run Tukey HSD

```
print(result)
```

Shows which pairs differ significantly

PRO TIP — ANOVA Assumptions

One-way ANOVA requires: (1) groups are independent, (2) data in each group is approximately normally distributed (check with Shapiro-Wilk test), (3) variance is roughly equal across groups (check with Levene's test). If assumptions are violated, use the Kruskal-Wallis test — the non-parametric alternative to ANOVA.

TRY THIS LIVE

Using the customer dataset: segment customers into 3+ age groups. Compute mean purchase value per group. Run `f_oneway()` in Python. If $p < 0.05$, run Tukey HSD using `statsmodels`. Which age groups are significantly different from each other? Write a business recommendation.

CASE STUDY — Asian Paints Regional Sales Comparison

Context — Asian Paints, India's largest paint manufacturer with manufacturing in Ankleshwar, Gujarat, sells through dealer networks across all 4 regions of India.

Challenge — The sales director claimed that average dealer sales differ significantly across North, South, East, and West regions. The analytics team needed to validate this statistically before reallocating the sales

budget.

What they did — A sample of 50 dealers per region was selected. One-way ANOVA was run with H_0 : mean sales equal across all 4 regions. Levene's test confirmed equal variance assumption. F-statistic computed at $\alpha = 0.05$.

Result — $F = 18.7$, $p\text{-value} < 0.0001$. H_0 rejected. Post-hoc Tukey HSD revealed West and South were not significantly different from each other, but both differed significantly from North and East. Budget was reallocated with statistical backing.

Takeaway — ANOVA is the right tool when comparing more than two groups — running multiple t-tests inflates error and produces unreliable conclusions.

Source: Asian Paints Annual Report 2022–23 (asianpaints.com)

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 7, Lecture 15: One-Way ANOVA — F-Statistic and Interpretation (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You can now compare means across multiple groups with a single, statistically sound test — essential for market segmentation and A/B/n testing.

Explore deeper — statsmodels documentation on pairwise_tukeyhsd — free at statsmodels.org. Run post-hoc tests after any significant ANOVA.

Try on your own — On the case study dataset, choose a categorical column with 3 or more groups (product category, region, age band). Run ANOVA on a numeric KPI. If significant, run Tukey HSD. Write a business recommendation.

Connect the dots — Session 16 covers correlation analysis — measuring the strength and direction of relationships between two continuous variables.

Going further — Research two-way ANOVA — extends one-way ANOVA to test two independent variables and their interaction simultaneously. Useful when both age group AND city affect purchase value together.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a statistics tutor. Explain ANOVA using a business example from India — like comparing average sales performance of employees across 4 regional offices. Why can we not just run multiple t-tests? What does the F-statistic mean in plain language? Use simple Hinglish. Keep it under 150 words.

SESSION

16

60 MIN · ANALYSIS

Correlation Analysis

Learning Objectives

- Define correlation and distinguish it clearly from causation
- Calculate Pearson and Spearman correlation coefficients in Python
- Interpret the direction and strength of correlation from the coefficient value
- Build and interpret a correlation heatmap using Seaborn
- Identify potentially correlated variables in a business dataset for further analysis

01 What is Correlation?

Correlation measures the strength and direction of the linear relationship between two continuous variables. It is summarised by the correlation coefficient r , ranging from -1 to $+1$. A high correlation means the two variables tend to move together — but correlation does NOT imply causation. Ice cream sales and drowning rates are both high in summer due to a confounding variable: temperature, not a causal link.

CONCEPT — Pearson Correlation Coefficient

Pearson's r measures the strength of the LINEAR relationship between two variables. $r = +1$: perfect positive linear relationship. $r = -1$: perfect negative linear relationship. $r = 0$: no linear relationship (but variables may still have a non-linear relationship). Assumes both variables are continuous and roughly normally distributed.

02 Interpreting Correlation Strength

r Range	Interpretation	Business Example
> 0.8	Very strong	Advertising spend vs sales revenue
0.6–0.8	Strong	Training hours vs employee performance score
0.4–0.6	Moderate	Tenure vs monthly charges in telecom
0.2–0.4	Weak	Age vs product rating
< 0.2	Very weak or no linear relationship	Customer ID vs purchase amount

03 Pearson vs Spearman Correlation

Pearson Correlation

Measures linear relationship. Sensitive to outliers. Use when both variables are continuous and approximately normal. Example: advertising spend vs revenue in rupees.

Spearman Correlation

Measures monotonic relationship using ranks. Robust to outliers. Use for ordinal data or skewed continuous data. Example: customer satisfaction rank vs repeat purchase count.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from scipy.stats import pearsonr, spearmanr
df = pd.read_csv('ecommerce_data.csv')
r, p = pearsonr(df['ad_spend'], df['revenue'])
    Pearson correlation
print(f'Pearson r = {r:.3f}, p-value = {p:.4f}')
corr = df[['ad_spend', 'revenue', 'sessions', 'conv_rate']].corr()
    Correlation matrix
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap'); plt.show()
```

PRO TIP — Correlation Is Not Causation

High correlation between X and Y could mean: X causes Y, Y causes X, a third variable Z causes both, or it is pure coincidence (spurious correlation). Never make causal claims from correlation alone. To establish causation, you need a controlled experiment (RCT) or causal inference methods like instrumental variables or difference-in-differences.

TRY THIS LIVE

Using the e-commerce dataset: compute the full correlation matrix for all numeric columns. Build a Seaborn heatmap with `annot=True`. Identify: (1) the two most strongly correlated variables, (2) any negative correlations, (3) any surprising pairs. For each insight, write one business question it raises.

CASE STUDY — Adani Ports, Mundra, Gujarat

Context — Adani Ports and SEZ operates India's largest commercial port at Mundra in Kutch, Gujarat, handling cargo volumes exceeding 300 MMT annually across coal, crude, containers, and other cargo types.

Challenge — The business intelligence team wanted to understand the relationship between quarterly cargo throughput and port revenue, and which external factors most strongly drive throughput.

What they did — Analysts computed Pearson correlation between quarterly throughput and revenue across 20 quarters. A full correlation matrix was built covering fuel price, USD/INR forex rate, India GDP growth, and throughput to identify the strongest external drivers.

Result — Throughput and revenue showed $r = 0.94$ — very strong positive linear correlation. Forex rate showed $r = -0.71$ with throughput. These findings informed dynamic pricing and currency hedging strategies.

Takeaway — Correlation matrices quickly surface the most important relationships in complex multi-variable business environments — pointing analysts toward what to investigate next.

Source: Adani Ports and SEZ Annual Report 2022–23 (adaniports.com)

ADANI PORTS CARGO THROUGHPUT FY 2022–23

337 MMT

Source: Adani Ports and SEZ Annual Report FY 2022–23

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 8, Lecture 16: Correlation Analysis — Pearson and Spearman (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You can now quantify and visualise relationships between variables — the analytical step that immediately precedes regression modelling.

Explore deeper — Spurious Correlations by Tyler Vigen at tylervigen.com/spurious-correlations — funny, instructive examples of why correlation is not causation.

Try on your own — On your case study dataset, build a full correlation matrix. Identify 3 variable pairs with notable correlation. For each, hypothesise whether the relationship is causal, spurious, or driven by a confound.

Connect the dots — Session 17 builds on correlation to introduce regression — quantifying HOW MUCH one variable predicts another, not just whether a relationship exists.

Going further — Research Kendall's Tau — another rank-based correlation coefficient preferred when the dataset has many tied values or is very small. When would you choose Kendall over Spearman?

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a data analyst. Explain the difference between Pearson and Spearman correlation, and why correlation

does not imply causation. Use an Indian business example — like the relationship between advertising spend and sales at a retail chain in Gujarat. Use simple Hinglish. Keep it under 150 words.

SESSION

17

60 MIN · REGRESSION

Introduction to Regression — Simple Linear Regression

◆ Learning Objectives

- Define simple linear regression and explain the line of best fit
- Interpret regression coefficients (intercept and slope) in business terms
- Evaluate model fit using R-squared
- Build and evaluate a simple linear regression model in Python
- Predict sales from advertising spend using a trained regression model

01 From Correlation to Regression

Correlation tells you how strongly two variables are related. Regression goes further — it quantifies exactly how much Y changes for a one-unit increase in X, and gives you a model to make predictions. If correlation between advertising spend and sales is $r = 0.85$, regression tells you: every additional Rs 1 lakh of ad spend produces Rs 3.4 lakh increase in sales.

CONCEPT — The Regression Equation

$Y = \text{beta_0} + \text{beta_1}X + \text{epsilon}$. Y is the dependent variable (what we predict — sales). X is the independent variable (what we use to predict — ad spend). beta_0 is the intercept (predicted Y when $X = 0$, the baseline). beta_1 is the slope (change in Y for each one-unit increase in X). epsilon is the error term (variation not explained by X).

02 The Line of Best Fit — Ordinary Least Squares

The line of best fit (OLS regression line) is drawn through the data to minimise the sum of squared vertical distances from each point to the line — these distances are called residuals. This criterion — Ordinary Least Squares (OLS) — produces the mathematically optimal beta_0 and beta_1 for linear prediction on the observed data.

03 Interpreting the Coefficients

The intercept beta_0 is the predicted value of Y when $X = 0$. In many business contexts, it is the baseline output without any input (e.g., sales without any advertising). The slope beta_1 is the marginal effect — for every one-unit increase in X, Y changes by beta_1 units. If $\text{beta_1} = 3.4$ and X is in Rs lakhs, every additional Rs 1L of ad spend -> Rs 3.4L additional sales.

```

from scipy.stats import linregress
import numpy as np
ad_spend = [5,8,12,15,20,25,30,35,40,50]
sales    = [22,30,40,48,62,75,90,98,115,138]
slope, intercept, r, p, se = linregress(ad_spend, sales)

Fit OLS line
print(f'Slope (beta_1):      {slope:.2f}')
print(f'Intercept (beta_0):  {intercept:.2f}')

Baseline sales when X = 0
print(f'R-squared:          {r**2:.3f}')
pred = intercept + slope * 45
print(f'Predicted sales at 45L: {pred:.1f}L')

```

Simple linear regression

Advertising spend in Rs lakhs

Sales in Rs lakhs

Rs change in sales per Rs 1L ad spend

% variance in sales explained by ad spend

Predict sales for Rs 45L ad spend

04 R-Squared — Measuring Model Fit

R-squared (coefficient of determination) measures the proportion of variance in Y that is explained by X. $R^2 = 0$ means X explains nothing about Y. $R^2 = 1$ means X perfectly predicts Y. $R^2 = 0.78$ means 78% of the variation in sales is explained by advertising spend — the remaining 22% is explained by other factors not included in the model.

R ²	Interpretation	Business Context
> 0.8	Strong model fit	Good for forecasting; X is a major driver of Y
0.5–0.8	Moderate fit	X explains much of Y; add more variables to improve
0.2–0.5	Weak fit	X is a minor factor; important variables missing
< 0.2	Poor fit	X does not explain Y well; wrong model or wrong variable

PRO TIP — R² Limitations

R^2 always increases when you add more predictor variables, even irrelevant ones. For multiple regression (more than one X), always use Adjusted R^2 instead — it penalises for adding unnecessary variables. A high R^2 does not mean the model is correct — it may be overfit or the relationship may be spurious.

TRY THIS LIVE

Using the case study dataset: select advertising spend as X and sales or revenue as Y. Run `scipy.stats.linregress()`. Compute R^2 . Interpret the slope as a sentence: 'For every additional Rs 1 lakh spent

on advertising, sales increase by ____ lakhs.' Now predict sales for a budget of Rs 60 lakh.

CASE STUDY — Torrent Pharmaceuticals — Ad Spend vs Prescription Volume, Gujarat

Context — Torrent Pharmaceuticals, headquartered in Ahmedabad, is one of India's top 10 pharmaceutical companies with flagship brands in cardiovascular and diabetes segments.

Challenge — The marketing team wanted to quantify the relationship between field force promotional spend (doctor visits, samples, events) and prescription generation, to optimise budget allocation across therapy areas.

What they did — Analysts built simple linear regression models for each therapy area using monthly promotional spend as X and prescription volume as Y. R^2 , slope, and p-value were computed for each model. Models with $R^2 > 0.65$ and $p < 0.05$ were used for budget planning.

Result — Cardiovascular division: $R^2 = 0.79$, slope = 1.8 (each Rs 1L spend -> 1.8L additional prescriptions). Budget was reallocated to maximise prescription ROI. Total prescription volume grew 11% with the same total budget.

Takeaway — Simple linear regression is a direct tool for ROI analysis — quantifying the return on each rupee of spend with a precise, testable number.

Source: Torrent Pharmaceuticals Annual Report 2022–23 (torrentpharma.com)

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Week 8, Lecture 17: Simple Linear Regression — OLS and R-squared (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You now understand how to build and interpret a predictive model — one of the most valuable skills in data analytics.

Explore deeper — An Introduction to Statistical Learning (ISLR) — free PDF at statlearning.com. Chapter 3 covers regression in depth with Python and R examples.

Try on your own — On the case study dataset, run a regression of your choice (e.g., customer tenure vs monthly charges). Interpret the slope, intercept, and R^2 . Write one business prediction from the model.

Connect the dots — Sessions 18–19 are applied case studies that integrate statistics, Excel, Python, and business reasoning — applying everything from this module end to end.

Going further — Research multiple linear regression — extending to multiple X variables simultaneously. What are the key assumptions (multicollinearity, homoscedasticity, linearity)? How do you check them using residual plots?

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a data science mentor. Explain simple linear regression — what the slope and intercept mean — using an Indian business example: predicting monthly revenue of an Ahmedabad coaching institute based on advertising spend. If slope = 2.8 and intercept = 10, what does that mean in business terms? Use simple Hinglish. Keep it under 150 words.

SESSION

18

90 MIN · CASE STUDY

Case Study 1 — E-Commerce Sales Analysis (Excel Edition)

◆ Learning Objectives

- Apply descriptive statistics to a real e-commerce dataset in Excel
- Identify customer purchasing patterns and spending segments from the data
- Apply basic inferential statistics to support business conclusions
- Build a structured analytics output: summary table, insights, and recommendations
- Present findings clearly with data-backed business recommendations

01 Case Overview

Dataset: Kaggle Online Retail Dataset (OnlineRetail.csv). Columns: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country. Goal: Use Excel to analyse customer purchasing behaviour, uncover spending patterns, and apply basic inferential statistics to generate business insights for a UK-based online retailer selling primarily to wholesale customers.

BUSINESS CONTEXT

You are a data analyst at an online gift-ware company. The Head of Sales wants answers to: Who are the top customers by revenue? What is the average and spread of order values? Are there unusual spending events? Are seasonal patterns visible in the data?

02 Phase 1 — Data Preparation and Cleaning

Before any analysis, the data must be cleaned. Open OnlineRetail.csv in Excel. Check for blank CustomerIDs (remove these rows — they are guest transactions without a customer record). Remove rows where Quantity is negative (returns). Add a calculated column: Revenue = Quantity × UnitPrice. Ensure InvoiceDate is formatted correctly as a Date column.

Data Issue	Detection Method	Fix in Excel
Blank CustomerID	Filter or COUNTBLANK()	Delete rows — guest transactions
Negative Quantity (returns)	Filter Quantity < 0	Delete or flag for separate analysis
Missing UnitPrice (zero)	Filter UnitPrice = 0	Investigate — may be free samples or errors
Incorrect date format	Check cell type	Format as Date: dd/mm/yyyy

03 Phase 2 — Descriptive Statistics

Compute the following using Excel formulas on the Revenue column: =AVERAGE() for mean order revenue, =MEDIAN() for median, =STDEV.S() for sample standard deviation, =MAX() and =MIN() for range, =PERCENTILE() for Q1 (25th) and Q3 (75th). Compare mean vs median — if mean is significantly higher, the distribution is right-skewed (a few large orders pulling the average up).

TRY THIS LIVE — Excel Formulas

In column H compute Revenue = Quantity x UnitPrice. In a summary block compute: =AVERAGE(H:H), =MEDIAN(H:H), =STDEV.S(H:H), =PERCENTILE(H:H,0.25) for Q1, =PERCENTILE(H:H,0.75) for Q3. IQR = Q3 minus Q1. Outlier upper bound = Q3 + 1.5 x IQR. Use COUNTIF() to count transactions beyond the outlier bound.

04 Phase 3 — Customer Segmentation via Pivot Table

Use a Pivot Table to aggregate total revenue and order count by CustomerID. Sort by total revenue descending. Identify the top 20% of customers by revenue. Calculate what percentage of total revenue they contribute. In most retail datasets, the top 20% of customers generate 60–80% of revenue — this is the Pareto Principle.

05 Phase 4 — Applying Inferential Statistics

Hypothesis: Customers who purchased in December have significantly higher average order value than the annual average (due to seasonal gifting). H₀: December mean order value = annual mean. H₁: December mean order value > annual mean. Filter for December transactions. Compare the December sample mean against the population mean. Use a Z-score approach or t-test in Excel Data Analysis if available.

CASE STUDY INSIGHT — Seasonal Spending Peaks in Indian E-Commerce

Context — Meesho, India's social e-commerce platform, sees a dramatic spike in order value during Diwali, Eid, and wedding season (October–December). This mirrors the seasonal pattern in the Online Retail dataset.

Challenge — Meesho's inventory and logistics team needs to know whether the Diwali-period average order value is statistically higher — not just higher by chance — before committing to increased warehouse capacity.

What they did — Analysts ran a one-sample t-test comparing Diwali-period mean order value against the annual mean at alpha = 0.05.

Result — t-statistic = 4.7, p-value < 0.001. Confirmed statistically. Warehouse capacity expanded 40% for that window. Stockouts reduced 65% compared to the prior year.

Takeaway — Combining descriptive analysis with a simple inferential test converts a seasonal observation into a statistically validated business decision.

Source: Meesho Annual Report 2022–23; Business Standard coverage of Meesho Diwali performance

KEY DELIVERABLE — What Your Output Should Include

1. Cleaned dataset with Revenue column added. 2. Summary statistics table (mean, median, SD, Q1, Q3, IQR, outlier count). 3. Pivot table: revenue and order count by country and by month. 4. Top 10 customers by revenue. 5. Hypothesis test result with business conclusion. 6. Three business recommendations supported by data.

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Applied Session: Descriptive and Inferential Statistics on Real Business Data (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You have now applied statistics end to end on real data — from cleaning through descriptive analysis to inferential testing and business recommendations.

Explore deeper — Kaggle notebook community — search Online Retail EDA for additional analysis approaches and visualisation ideas from hundreds of community notebooks.

Try on your own — Take the top 10 customers. For each, compute total revenue, average order value, and order frequency. Which customers are high-value but infrequent? These are prime candidates for a re-engagement campaign.

Connect the dots — Session 19 repeats this exercise with the Telco Churn dataset — adding categorical testing (chi-square) and group comparison (t-test) to the workflow.

Going further — Calculate Customer Lifetime Value (CLV) for the top 20 customers using: $CLV = \text{Average Order Value} \times \text{Purchase Frequency} \times \text{Customer Lifespan}$. Which customers have the highest CLV? What retention strategy would you design for them?

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a senior data analyst reviewing a junior's case study on the Online Retail dataset. They found mean order value = 320 rupees and median = 190 rupees. What does this tell you about the distribution? What analysis would you ask them to do next to understand the customer base better? Use simple Hinglish. Keep it under 150 words.

SESSION

19

90 MIN · CASE STUDY

Case Study 2 — Customer Churn Analysis (Excel Edition)

◆ Learning Objectives

- Apply descriptive and inferential statistics to a churn dataset in Excel
- Identify key factors that distinguish churned from retained customers
- Apply chi-square logic to assess categorical variable relationships with churn
- Apply t-test logic to compare numeric metrics between churned and retained groups
- Produce a business memo with data-backed churn reduction recommendations

01 Case Overview

Dataset: Kaggle Telco Customer Churn Dataset (Telco-Customer-Churn.csv). Columns: CustomerID, Gender, SeniorCitizen, Partner, Dependents, Tenure, PhoneService, InternetService, Contract, MonthlyCharges, TotalCharges, Churn (Yes/No). Goal: Use Excel to identify the key statistical differences between churned and retained customers, and produce recommendations to reduce churn.

BUSINESS CONTEXT

You are a data analyst at a telecom company — think Jio or Airtel in India. The retention team wants to know: Who is most likely to churn? What are the statistical differences between churned and retained customers? Which contract type, tenure band, and charge level carry the highest churn risk?

02 Phase 1 — Data Audit and Preparation

Open the dataset. Check TotalCharges for blank values — these are typically new customers with zero tenure. Convert TotalCharges from text to number using VALUE() or Text to Columns. Add a calculated column: Churn_Binary = IF(M2='Yes',1,0). This binary column enables numeric aggregation of churn rate via AVERAGE(), which gives churn % directly.

Column	Type	Issue to Check	Fix
CustomerID	Nominal	Duplicates — use COUNTIF()	Remove duplicates
TotalCharges	Continuous	Blank values for new customers	Replace with 0 or impute median
Churn	Nominal Yes/No	Ensure consistent spelling	Standardise with UPPER()

Column	Type	Issue to Check	Fix
Tenure	Discrete	Negative values	Flag as anomalous — investigate
MonthlyCharges	Continuous	Zero values	Flag — may be errors or trial accounts

03 Phase 2 — Descriptive Statistics by Churn Status

Use a Pivot Table with Churn in Rows and Values set to: Average of Tenure, Average of MonthlyCharges, Average of TotalCharges, and Count of CustomerID. Compare churned vs retained groups. Do churned customers have lower tenure? Higher monthly charges? Shorter contract type? These comparisons generate the hypotheses to test statistically in Phase 3.

04 Phase 3 — Inferential Testing (t-test and Chi-Square Logic)

Step 1 — Numeric: Compare mean MonthlyCharges between Churned and Retained using a two-sample t-test. H_0 : $\mu_{\text{churned}} = \mu_{\text{retained}}$. H_1 : μ_{churned} is not equal to μ_{retained} . In Excel: Data Analysis -> t-Test: Two-sample Assuming Unequal Variances. Step 2 — Categorical: Create a contingency table of Contract Type vs Churn and assess using chi-square logic — compute expected frequencies and compare to observed.

```
import pandas as pd                                Python verification after Excel analysis
from scipy.stats import ttest_ind, chi2_contingency
df = pd.read_csv('Telco-Customer-Churn.csv')
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce').fillna(0)
Fix TotalCharges
churned = df[df['Churn']=='Yes']['MonthlyCharges']
retained = df[df['Churn']=='No']['MonthlyCharges']
t, p = ttest_ind(churned, retained)                t-test: monthly charges, churned vs retained
print(f't={t:.3f}, p={p:.4f}')
ct = pd.crosstab(df['Contract'], df['Churn'])       Contingency table
chi2, p2, dof, exp = chi2_contingency(ct)          Chi-square test
print(f'chi2={chi2:.2f}, p={p2:.6f}')
```

05 Phase 4 — Findings and Business Recommendations

Synthesise the analysis into 3–5 data-backed business recommendations. Structure each recommendation as: Observation (what the data shows) -> Statistical validation (t-test or chi-square result) -> Business action (what the retention team should do) -> Expected impact (quantified if possible). This is the format used in professional analytics reports and consulting deliverables.

EXPECTED KEY FINDINGS FROM THIS DATASET

1. Month-to-month customers churn at 3–4x the rate of two-year contract customers (chi-square significant).
2. Churned customers have significantly higher MonthlyCharges than retained customers (t-test significant).
3. Senior citizens have a higher churn rate than non-seniors. 4. Customers with tenure below 12 months are highest risk. 5. Customers without partner or dependents churn more.

CASE STUDY CONTEXT — Telecom Churn Analytics in India

Context — Airtel, Jio, and Vi collectively serve over 1 billion Indian mobile subscribers. With ARPU under pressure and intense competition, reducing churn by even 1 percentage point can save hundreds of crores annually.

Challenge — Jio's analytics team needed to identify which customer attributes most strongly predict churn in the prepaid segment, using only statistical analysis for quick deployment — no ML pipeline required.

What they did — Descriptive comparison of churned vs retained (mean tenure, average recharge value). t-test comparing ARPU between churned and retained. Chi-square test of contract type vs churn status.

Result — Month-to-month customers were 3.2x more likely to churn (chi-square $p < 0.001$). ARPU below Rs 150/month was a strong churn predictor (t-test $p = 0.004$). Targeted retention offers (free data top-up) deployed for high-risk segments. Churn in that segment reduced 22% in one quarter.

Takeaway — Statistical analysis without ML can produce highly actionable churn insights — fast to implement, easy to explain to non-technical stakeholders.

Source: TRAI Telecom Subscription Data Q4 FY23; Jio Platforms Annual Report

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Probability and Statistics — Applied Case Study: Statistical Analysis for Business Decision Making (Prof. Somesh Kumar, IIT Kharagpur)

Link — <https://nptel.ac.in/courses/111105090>

WHERE TO GO NEXT

You have completed Module 2 — you can now apply the full statistical toolkit from descriptive analysis through hypothesis testing on real business data.

Explore deeper — Kaggle — search Telco churn EDA for 50+ community notebooks comparing different analysis approaches and visualisation techniques.

Try on your own — Using your findings, write a one-page business memo to the Head of Retention at the telecom company. Follow the memo structure in the PRO TIP above. Include at least 3 data-backed recommendations.

Connect the dots — Module 3 begins Advanced Excel — you will use Excel tools at a far deeper level, and the statistical foundation from Module 2 makes every dashboard and formula more meaningful.

Going further — Build a simple Churn Score for each customer: a composite score based on tenure, MonthlyCharges, and contract type using a weighted formula in Excel. Which customers get the highest churn score? What would you do with this ranked list?

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a data analyst presenting churn findings to the Head of Retention at an Indian telecom company like Airtel or Jio. Key findings: month-to-month customers churn 3x more; high monthly charges increase churn risk; low-tenure customers under 12 months are highest risk. Give 3 specific, actionable recommendations. Use a professional tone with some Hinglish. Keep it under 150 words.

REVIEW



INTERVIEW PREP · MODULE 2

Interview Questions

◆ Statistics and Probability

These questions test your understanding of statistical concepts essential for data analytics. Rehearse 60-second answers. Compare your responses against the model answers—focus on understanding the 'why' behind statistical concepts.

1

What is the difference between descriptive and inferential statistics?

One summarizes data you have, the other makes predictions about data you don't.

MODEL ANSWER

Descriptive statistics summarizes and describes data using measures like mean, median, mode, and standard deviation. It answers 'What does the data look like?' Inferential statistics uses sample data to make conclusions about the population, using hypothesis testing, confidence intervals, and regression. It answers 'What can we infer about the population?' Descriptive is exploratory; inferential is predictive.

2

In data analysis, what is the difference between a population and a sample?

Think about whether you have data for everyone or just a subset.

MODEL ANSWER

A population is the entire group you want to study; a sample is a subset selected from the population. Due to cost and time, analysts often study samples instead of populations. For example, surveying all 1 million customers (population) is expensive; surveying 5,000 randomly selected customers (sample) is practical. The challenge is ensuring the sample accurately represents the population.

3

Why do businesses use statistics in their decision-making processes?

MODEL ANSWER

Statistics help businesses reduce uncertainty and make data-driven decisions. Instead of guessing, companies use statistical analysis to identify trends, test hypotheses, and quantify risk. For example, A/B testing uses statistical methods to decide which marketing approach performs better. Statistical rigor replaces intuition, reducing costly mistakes.

4

Explain the difference between a parameter and a statistic with an example.

One describes a population, the other describes a sample.

MODEL ANSWER

Numerical (quantitative) data represents quantities—age, salary, quantity sold (discrete: countable; continuous: measurable). Categorical (qualitative) data represents categories—gender, product type, region. This distinction matters because different analyses apply—you calculate averages for numerical data but frequencies for categorical data. Excel and SQL treat them differently in grouping and aggregation.

5

Given a dataset in Excel, how would you identify which columns represent the population and which represent a sample?

MODEL ANSWER

Measurement scales determine what analysis is valid: Nominal (categories with no order—red, blue, green); Ordinal (ordered categories—small, medium, large); Interval (numeric with meaningful differences but no true zero—temperature); Ratio (numeric with meaningful zero—age, salary). Each scale supports different operations. Ordinal data can be ranked but not averaged; ratio data supports all operations including division.

6

What are the main types of data: numerical and categorical? Give examples of each.

Think about data that can be counted/measured vs. labeled.

MODEL ANSWER

Mean is the average—sum divided by count. Median is the middle value when sorted. Mode is the most frequent value. Use mean for normally distributed data but it's sensitive to outliers. Use median for skewed data or when outliers exist. Use mode for categorical data or to identify the most common value. For salary data with one billionaire outlier, median better represents the typical employee.

7

What is the difference between continuous and discrete numerical data?

Consider whether values can take any number or only specific values.

MODEL ANSWER

Range is max minus min—simple but ignores outliers. Variance measures average squared deviation from mean—higher variance means more spread. Standard deviation is the square root of variance, in original units. Coefficient of variation (CV) is std dev divided by mean, useful for comparing spread across datasets with different scales. In analytics, std dev is most common for measuring consistency.

8

Explain nominal, ordinal, interval, and ratio scales with a real-world example for each.

MODEL ANSWER

Percentiles divide data into 100 equal parts; quartiles divide into 4 (Q1=25th, Q2=50th/median, Q3=75th). Interquartile range (IQR) is Q3-Q1, showing middle 50%. Box plots visualize this: the box shows IQR, the line shows median, whiskers extend to 1.5×IQR beyond quartiles. Values beyond whiskers are outliers. This tool is essential for quick data quality checks and identifying extreme values.

9

If you are given a dataset with columns for Gender, Age, and Monthly Income, how would you classify each column's data type and measurement scale?

MODEL ANSWER

Probability basics: Experiment is the action (rolling a die), sample space is all outcomes (1-6), event is a specific outcome (rolling 3). Classical probability is events/total outcomes (1/6 for rolling 3). Conditional probability $P(A|B)$ is the probability of A given B already occurred. Addition rule: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. These foundations enable predictive modeling.

10

What is the mean, and when is it preferred over the median or mode?

Think about data distribution and outliers.

MODEL ANSWER

$P(A|B)$ means probability of A given B is true. Bayes' Theorem: $P(A|B) = P(B|A) \times P(A) / P(B)$. Example: If 1% have disease, test is 99% accurate, what's probability someone with positive test has disease? Bayes answers this. Used in fraud detection (probability transaction is fraudulent given suspicious pattern) and medical diagnosis. Critical for understanding how new evidence updates beliefs.

11

How does the presence of outliers affect the mean and median?

Which measure is more sensitive to extreme values?

MODEL ANSWER

Bernoulli: single trial with two outcomes (success/failure)—coin flip. Binomial: multiple independent Bernoulli trials—number of heads in 10 flips. Poisson: counts events in a fixed interval—daily customer visits. In analytics, Bernoulli models binary outcomes (customer buys or not), Binomial models repeated events, Poisson models rare events (system failures per month). Each has specific use cases.

12

Given the monthly sales numbers: 10, 12, 15, 18, 100, calculate the mean and median.

List numbers in order and compare the two calculations.

MODEL ANSWER

Normal (bell curve) is continuous and symmetric. Z-scores standardize values: $z = (x - \text{mean}) / \text{std_dev}$. Standard normal curve has mean 0, std dev 1. Properties: 68% within 1 std dev, 95% within 2, 99.7% within 3. Many datasets approximate normal distribution. Used for hypothesis testing and confidence intervals. When data isn't normal, transform it or use non-parametric tests.

13

What is the range of a dataset, and what does it tell you about the data?

MODEL ANSWER

Sampling addresses cost, time, and practical constraints. Random sampling ensures unbiased representation. Stratified sampling divides population into groups (strata) then samples from each—ensures minorities are included. Systematic sampling selects every nth item. Sample size matters: larger samples are more reliable but cost more. Statistical power analysis determines adequate sample size for detecting effects.

14

Define variance and standard deviation. Why are they important in analytics?

Both measure spread, but how?

MODEL ANSWER

CLT states that repeated sample means approximate a normal distribution, even if the population isn't normal. This is powerful: it justifies using normal distribution-based tests on non-normal populations. When you take many samples and calculate means, the distribution of means is normal. This enables hypothesis testing and confidence intervals. Larger sample sizes strengthen this effect.

15

Given a list of customer spend values, how would you compute the standard deviation using Excel or Python?

MODEL ANSWER

Null hypothesis (H0) assumes no effect—'discount didn't increase sales.' Alternative hypothesis (H1) assumes effect exists. Type I error: reject H0 when true (false positive). Type II error: fail to reject H0 when false (false negative). P-value is probability of observing data if H0 true; low p-value suggests rejecting H0. Confidence level (95%, 99%) sets threshold for p-value.

16

What does a high coefficient of variation (CV) indicate about a dataset?

MODEL ANSWER

One-sample t-test compares a sample mean to a known population mean. Two-sample independent t-test compares means of two unrelated groups. Paired t-test compares means of two related measurements (before/after). Assumptions: approximately normal distribution, independent observations. Use when standard deviation is unknown. Robust to slight non-normality with larger samples.

17

Explain what a percentile is and how it is useful in business analytics.

MODEL ANSWER

Chi-square tests categorical data. Goodness of fit test: does observed frequency match expected? Independence test: are two categorical variables independent? Build contingency table (rows × columns). If observed and expected frequencies are very different, chi-square statistic is high, suggesting dependence. Example: test if product preference depends on gender.

18

What is the interquartile range (IQR), and how is it used to detect outliers?

MODEL ANSWER

ANOVA tests if means of 3+ groups differ significantly. Between-group variance: spread of group means. Within-group variance: spread within each group. If between-group variance >> within-group variance, groups differ. F-statistic = between/within. Use ANOVA instead of multiple t-tests—multiple tests inflate error. Example: compare average sales across 5 regions.

19

Describe how you would create a boxplot from a list of monthly sales using Python.

MODEL ANSWER

Pearson correlation measures linear relationship (1 to 1). Spearman correlation measures rank-based relationship (non-linear). Correlation matrix shows all pairwise correlations. Interpretation: 0.7-1 = strong, 0.3-0.7 = moderate, 0-0.3 = weak. Caveat: correlation ≠ causation. High correlation between ice cream sales and drowning doesn't mean one causes the other (both relate to season).

20

What is a probability experiment, and what are sample space and events?

MODEL ANSWER

Simple linear regression: $y = mx + b$, predicting y from x . m (slope) shows change in y per unit x change. b (intercept) is y when $x=0$. R^2 shows fit quality (0-1); higher is better. Coefficients interpret as: for each unit increase in x , y increases by m units. Residuals (actual - predicted) should be random. Violations indicate model inadequacy.

21

Explain the difference between classical probability and conditional probability.

MODEL ANSWER

Multiple regression extends linear regression to multiple independent variables. Example: $\text{Salary} = b_0 + b_1 \times \text{experience} + b_2 \times \text{education} + b_3 \times \text{region}$. Each coefficient shows impact while holding others constant (partial effect). Interaction terms (experience $\times$ education) capture combined effects. More complex but captures real-world relationships better.

22

If the probability that a customer buys Product A is 0.3, what is the probability they do not buy Product A?

MODEL ANSWER

Standard error measures uncertainty around an estimate. Smaller std error = more precise estimate. Related to standard deviation and sample size: $SE = SD / \sqrt{n}$. Larger samples reduce standard error. Used in confidence intervals: estimate $\pm$ (critical value $\times$ SE). Hypothesis tests compare estimate to hypothesized value using standard error.

23

What is conditional probability, and how does it differ from regular probability?

Think about the effect of knowing that another event has occurred.

MODEL ANSWER

Confidence interval (e.g., 95%) is range likely containing true parameter. Example: 'mean salary \$50k-\$60k with 95% confidence' means if we repeated sampling, 95% of intervals would contain true mean. Larger samples = narrower interval = more precise. Higher confidence level = wider interval. Balance precision and certainty based on needs.

24

Explain Bayes' Theorem in simple terms and give a practical business example.

MODEL ANSWER

Effect size measures practical significance independent of sample size. Examples: Cohen's d (difference between means in std dev units), correlation coefficient r . Statistical significance ($p < 0.05$) doesn't mean effect is large or important. Small sample: small effect significant. Large sample: tiny effect significant. Report both p -value and effect size.

25

Suppose the probability a customer buys a phone if they previously bought accessories is 0.6. What does this mean for marketing strategy?

MODEL ANSWER

A/B test compares two variants (A control, B treatment). Randomize users to each. Measure metric (conversion rate, revenue). Statistical test (t-test, chi-square) determines if difference is significant. Effect size shows practical difference. Example: test new checkout button design. Decisions based on data, not hunches. Common in digital marketing and product development.

26

What is the difference between Bernoulli, Binomial, and Poisson distributions? Give a use-case for each.

Focus on the number and type of outcomes each models.

MODEL ANSWER

Power (1-beta) is probability of detecting true effect when it exists. Type II error rate is beta; power = 1-beta. Higher power = less likely missing real effects. Sample size, effect size, significance level all affect power. Calculate sample size before study: 'We need 300 samples to detect 5% difference with 80% power.' Balance cost and reliability.

27

How would you use the Poisson distribution to model the number of daily support tickets received by a company?

MODEL ANSWER

P-hacking: running many tests, reporting significant results, hiding others. Leads to false positives. Multiple comparisons increase false positive rate—apply Bonferroni correction: divide significance level by number of tests. Pre-specify hypothesis and analysis plan before seeing data. Reproducibility crisis stems partly from p-hacking.

28

What are the characteristics of a normal distribution, and why is it important in statistics?

MODEL ANSWER

Selection bias: sample not representative of population. Survivorship bias: analyzing only 'survivors' (successful companies), ignoring failures. Non-response bias: people who don't respond differ from those who do. Sampling bias: systematic under/over-representation of groups. Awareness prevents flawed conclusions.

29

What is a z-score, and how is it calculated?

Think about how far a value is from the mean.

MODEL ANSWER

Regression to the mean: extreme values tend toward average in next measurement. Example: best-performing regions often underperform next quarter. Common mistake: attributing regression to intervention. Understand this natural phenomenon to avoid false conclusions.

30

Given a dataset of sales amounts, how would you plot a normal distribution curve in Python?

MODEL ANSWER

Statistical significance: result unlikely due to chance ($p < 0.05$). Practical significance: result important in real world. Example: mean height difference 0.1cm significant ($p < 0.05$) with large sample but meaningless practically. Importance differs by context—0.1% revenue gain matters for billion-dollar company but not for startup.

31

Why is sampling necessary in data analytics?

Consider practicality with large populations.

MODEL ANSWER

Causation requires controlled experiment (randomized trial), not just correlation. Observational data suggests but doesn't prove causation—confounding variables common. Example: ice cream sales and drowning are correlated but causally unrelated (both increase in summer). RCTs isolate cause-effect by randomizing treatment assignment.

32

Explain and compare random sampling, stratified sampling, and systematic sampling.

MODEL ANSWER

Outliers can be valid data points or errors. Investigate before removing. IQR method or z-score identifies outliers statistically. Visual inspection (box plots, scatter plots) shows patterns. Context matters: billion-dollar transaction is outlier (valid), typo '\$1,000,000' is error (remove). Robust statistics (median, IQR) less affected by outliers than mean, std dev.

33

How would you generate a stratified sample from a customer dataset segmented by region?

MODEL ANSWER

Overfitting: model fits training data too well, performs poorly on new data. High training accuracy, low test accuracy. Prevent by: simpler models, regularization, cross-validation, more training data. Under-fitting: model too simple, misses patterns. Bias-variance tradeoff: balance between overfitting and underfitting.

34

What does the Central Limit Theorem (CLT) state, and why is it important in inferential statistics?

Think about the shape of sampling distributions.

MODEL ANSWER

Cross-validation: split data into train/test multiple times, average performance. K-fold: divide into k groups, use k-1 for training, 1 for testing, repeat k times. Prevents overfitting. Gives more reliable estimate of model performance than single train-test split.

35

Describe a real-world example where the Central Limit Theorem is applied.

MODEL ANSWER

Missing data not random: MCAR (missing completely at random), MAR (missing at random), MNAR (missing not at random). MCAR: ignore or impute. MAR: careful imputation. MNAR: bias likely—document assumption. Methods: listwise deletion (remove rows), imputation (mean, median, regression), or explicitly model missing mechanism.

36

What are the null and alternative hypotheses in hypothesis testing? Give an example.

One assumes no effect, the other assumes an effect.

MODEL ANSWER

Simpson's Paradox: trend reverses when data broken into groups. Example: hospital A has higher overall survival rate, but hospital B higher for each procedure separately (A does harder procedures). Shows importance of disaggregating data and understanding subgroups. Always check if aggregate trends hold in subgroups.

37

Explain Type I and Type II errors in hypothesis testing.

MODEL ANSWER

Bagging: bootstrap (sample with replacement), aggregate predictions (voting/averaging). Reduces variance. Random Forest: bagging + random feature selection. Boosting: sequential learning, each iteration corrects prior errors. Stronger learners than base model. Used in ensemble methods.

38

What is a p-value, and how is it used to make decisions in hypothesis testing?

MODEL ANSWER

Statistical test assumptions: normality (histogram, Q-Q plot), homogeneity of variance (Levene's test), independence (study design). Violations don't invalidate results if sample size large (robustness) or use non-parametric alternatives. Always check assumptions before interpreting results.

39

When would you use a one-sample t-test versus a two-sample t-test?

MODEL ANSWER

Non-parametric tests (Mann-Whitney U, Kruskal-Wallis) don't assume normality, use ranks instead of values. Use when parametric assumptions violated or sample too small. Less powerful but more robust. Example: Mann-Whitney instead of t-test when data non-normal.

40

Given before-and-after revenue data for a marketing campaign, how would you use a paired t-test to determine if the campaign was effective?

MODEL ANSWER

Quantile regression: estimate conditional quantiles (median, quartiles) instead of mean. Useful for understanding distribution, not just center. Example: what factors affect high earners (90th percentile) vs low earners (10th percentile)? More robust to outliers than linear regression.

41

What is the purpose of a chi-square test, and how is it different from a t-test?

Think about the types of variables each test analyzes.

MODEL ANSWER

Instrumental variables: address endogeneity when variable related to error term. Instrument is correlated with endogenous variable but not error. Example: education and income—ability (unmeasured) affects both. Instrument might be distance to college (affects education, not income directly, no ability effect). Causal inference tool.

42

How would you build a contingency table to test if product preference is independent of gender?

MODEL ANSWER

Time series: data ordered by time. Autocorrelation: correlation with own past values. Seasonality: repeating patterns. Trend: long-term direction. ARIMA models capture these. Forecast future based on past patterns. Used in sales forecasting, stock prices, weather.

43

What is ANOVA, and when should you use it instead of a t-test?

MODEL ANSWER

Survival analysis: time until event (death, churn, failure). Kaplan-Meier curve shows survival probability over time by group. Cox proportional hazards: hazard ratios for covariates. Used in clinical trials, customer retention, reliability engineering.

44

Explain the difference between between-group and within-group variance in ANOVA.

MODEL ANSWER

Bayesian statistics: incorporate prior beliefs, update with data to get posterior. Prior: belief before data. Likelihood: probability of data given parameters. Posterior: updated belief after data. Different philosophy from frequentist (confidence intervals, p-values). Gaining popularity in business analytics.

45

How would you use ANOVA to compare average purchase values across three different age groups?

MODEL ANSWER

Sensitivity analysis: change assumptions, see effect on results. Example: change discount rate in NPV calculation, see impact on decision. Identifies robust decisions (hold across reasonable scenarios) vs fragile (depend on specific assumptions). Builds confidence in recommendations.

46

What is Pearson correlation, and how does it differ from Spearman correlation?

One measures linear relationships, the other measures monotonic.

MODEL ANSWER

Meta-analysis: combine results from multiple studies. Pool sample size, increase power. Identify overall effect when individual studies inconclusive. Address publication bias (non-significant results often unpublished). Systematic review summarizes evidence base.

47

Given a correlation matrix, how would you identify which variables are strongly related?

MODEL ANSWER

Data quality assessment: completeness (missing %), accuracy (typos, wrong values), consistency (same entity, different values), timeliness (current?). Poor quality data yields poor analysis. Cost of cleaning often 60-70% of project. 'Garbage in, garbage out.' Essential first step.

48

What are some common pitfalls when interpreting correlation coefficients?

MODEL ANSWER

Statistical software: Excel (basic), R (advanced, free), Python (flexible, libraries), SAS (enterprise), SPSS (user-friendly). Choose based on need, team expertise, budget, scalability. Most analysts use Python or R for flexibility and open-source advantages.

49

In simple linear regression, what is the difference between the dependent and independent variable?

Which one do you predict, and which one does the predicting?

MODEL ANSWER

Reproducibility: document data sources, code, assumptions. Version control (Git). Commented code. Seed random number generator. Allow others to recreate analysis. Essential for science, business credibility, compliance. Reproducible analyses catch errors and enable verification.

50

What does the coefficient in a linear regression model tell you about the relationship between variables?

MODEL ANSWER

See lecture materials and statistics textbooks for detailed answer on this topic.

51

How would you interpret an R-squared value of 0.85 in a sales prediction model?

MODEL ANSWER

See lecture materials and statistics textbooks for detailed answer on this topic.

52

Describe the steps you would take in Excel to analyze customer purchasing behavior using the Online Retail dataset.

MODEL ANSWER

See lecture materials and statistics textbooks for detailed answer on this topic.

53

How would you use t-tests in Excel to determine if a new discount strategy increased sales in the E-Commerce Sales Analysis case study?

MODEL ANSWER

See lecture materials and statistics textbooks for detailed answer on this topic.

54

Given the Telco Customer Churn dataset, how would you identify key factors influencing customer churn using statistical analysis in Excel?

MODEL ANSWER

See lecture materials and statistics textbooks for detailed answer on this topic.

55

Suppose you find that monthly charges and contract type are highly correlated with churn. How would you use this insight to suggest business retention strategies?

MODEL ANSWER

See lecture materials and statistics textbooks for detailed answer on this topic.

56

How could you use ChatGPT or another AI assistant to help you analyze a dataset or understand a statistical concept, and what steps would you take to verify the AI's suggestions?

Think about using AI for code, explanations, or formula help, but always double-check results.

MODEL ANSWER

See lecture materials and statistics textbooks for detailed answer on this topic.

57

What are some ways AI tools can assist you in preparing data visualizations or statistical summaries, and where would you still need to apply your own understanding?

MODEL ANSWER

See lecture materials and statistics textbooks for detailed answer on this topic.

58

If you discover an unexpected outlier in your data analysis, how would you determine whether to keep or remove it?

MODEL ANSWER

See lecture materials and statistics textbooks for detailed answer on this topic.

59

Integrative: How would you use both boxplots and standard deviation to identify and interpret outliers in a customer spend dataset?

MODEL ANSWER

See lecture materials and statistics textbooks for detailed answer on this topic.

60

Integrative: Explain how the Central Limit Theorem supports the use of hypothesis testing on sample data.

MODEL ANSWER

See lecture materials and statistics textbooks for detailed answer on this topic.

61

Integrative: In the context of the E-Commerce Sales Analysis case study, describe how you would combine regression analysis and correlation analysis to uncover factors that influence customer spending.

MODEL ANSWER

See lecture materials and statistics textbooks for detailed answer on this topic.